

Improving Precipitation Phase Classification in Mountain Hydrology: A Machine Learning Framework Using Citizen Science and Gridded Meteorological Data

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ABSTRACT: Precipitation phase partitioning is a leading source of uncertainty in mountain hydrology, where rain–snow misclassification degrades streamflow forecasting and water-resource management. Conventional methods relying on sparse station networks and near-surface thermodynamic thresholds are limited near freezing, where rain and snow distributions overlap. Our framework assimilates crowdsourced phase reports from the Mountain Rain or Snow (MRoS) network with station meteorology and satellite-derived probability of liquid precipitation, interpolates every source to an hourly 1-km grid, and trains a calibrated XGBoost machine learning model on held-out MRoS labels across California’s Sierra Nevada and Colorado’s Rocky Mountains from October 2022 to May 2026. We benchmark against nine standard partitioning methods with a paired cluster bootstrap. The model reached 91.2% accuracy in California and 91.4% in Colorado, and 85.6% and 82.7% near freezing against 69.1% and 74.3% for the best benchmark in each domain, with a test AUROC of 0.965 in both. Its accuracy gain over that benchmark was 11.1 percentage points in California (95% CI 9.0–13.4) and 16.5 points near freezing (12.2–21.1). Though Colorado’s accuracy gain was marginal (3.6 points, 0.2–6.9), its macro F1 gain was meaningful (0.096, 0.042–0.152). Rather than classify mixed precipitation as a distinct phase, we compare observer-reported mixed events against the model’s region of uncertainty, which flags where mixed phase is likely and captures about a third of individual events. Ablations trace these gains to the assimilated predictors, above all the crowdsourced surfaces.

SIGNIFICANCE STATEMENT: Predicting whether mountain precipitation falls as rain or snow is essential for water-supply and flood forecasting. However, the process is challenging and prone to errors near freezing temperatures, where sparse weather stations and simple temperature-based partitioning methods struggle to correctly classify precipitation phase. We show that combining crowdsourced phase reports from volunteer observers with station, satellite, and terrain data on a common 1-km grid substantially improves rain–snow prediction across California and Colorado, especially near freezing.

1. Introduction

Determining whether cold-season precipitation falls as rain or snow is a critical yet challenging area of research in mountain hydrology. Seasonal mountain snowpack supplies water to more than one-sixth of the global population (Barnett et al. 2005), and in the western United States 53% of total runoff originates as snowmelt, rising to roughly 70% in mountainous terrain (Li et al. 2017). That snowpack functions as natural storage, accumulating through the winter and releasing gradually during melt, whereas rain runs off within hours, so precipitation phase at the time of a storm determines which of the two occurs. A

single storm can therefore recharge water supply or trigger flooding depending on its phase, and an error in phase classification propagates downstream into snowpack estimates, runoff timing, flood magnitude, and the forecasts issued to mountain communities and backcountry users (Harpold et al. 2017; Jennings and Molotch 2019). Warming temperatures have meanwhile shifted more cold-season precipitation from snow to rain and increased rain-on-snow flood risk across western North America (Musselman et al. 2018), compounding that uncertainty for water supply planning and hazards forecasting.

Phase remains difficult to classify accurately, especially near freezing. Precipitation gauges in mountainous terrain are geographically sparse relative to the heterogeneous conditions they are intended to record. Few stations report phase at all, and those that do often sit where cold-air pooling, wind exposure, and steep elevation gradients decouple recorded estimates from true conditions in their surroundings (Lundquist et al. 2008; Minder et al. 2011; Wayand et al. 2016). Phase also resists a clean definition: near freezing, hydrometeors may arrive as graupel, ice pellets, freezing rain, or wet mixtures that share a range of surface air temperatures (Section d). This physical ambiguity makes classifying mixed precipitation separately from rain or snow both a predictive challenge and a significant source of error.

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Conventional partitioning methods struggle as well. Most land surface, hydrologic, and snow models partition rain from snow with a single near-surface air-temperature threshold or a smooth transition around one (Ding et al. 2014; Dai 2008). This approach measurably alters simulated accumulation and melt (Jennings and Molotch 2019) without translating to regions where transitional phase temperatures can vary by more than 4°C (Jennings et al. 2018). Wet-bulb temperature supplements phase classification by capturing the cooling of hydrometeors falling through unsaturated air (Harder and Pomeroy 2013; Sims and Liu 2015; Wang et al. 2019), but no surface measurement resolves melting and refreezing aloft, so errors concentrate near freezing (Feiccabrino et al. 2015). Increasing model complexity does not necessarily fix the issue either, since Jennings et al. (2025) found that fine-tuning machine learning models only marginally improves on simple partitioning rules given near-surface meteorology.

Addressing this ceiling in predictive accuracy requires new model inputs. Ground-based phase reports at transition-zone elevations can supplement synoptic weather stations, which are sparse and typically biased toward lower elevations and airports. Crowdsourcing helps fill atmospheric observations where instruments are absent (Muller et al. 2015), and the Mountain Rain or Snow (MRoS) project does so by collecting timestamped, geolocated visual reports from citizen scientist volunteers across mountain regions (Arienzo et al. 2021). Such reports have helped characterize rain–snow and mixed conditions and have been used to benchmark phase partitioning methods (Jennings et al. 2023, 2025). To our knowledge, their value as a predictor—assimilated alongside station and satellite data into a spatially continuous, probabilistic field—has yet to be explored.

This study explores how to overcome these limitations in predictive accuracy of phase. Three research questions frame the work:

1. Can assimilating a wider variety of predictors (i.e., interpolated crowdsourced phase observations, interpolated station meteorology, and satellite-derived probability of liquid precipitation (pLP)) improve rain–snow accuracy beyond what near-surface thermodynamics alone can supply?
2. Does this data assimilation approach enhance predictive phase accuracy near freezing where conventional threshold partitioning methods fail?
3. Can mixed precipitation, which is usually discarded or forced into an ill-defined class, exhibit meaningful signal in a model’s region of predictive uncertainty?

The framework we propose combines four complementary sources of data, each compensating for what the others

lack: crowdsourced phase reports from MRoS, hourly station meteorology, satellite pLP from GPM IMERG (Huffman et al. 2020), and topographic elevation (Section 3). The station meteorology and crowdsourced observations are interpolated into their own surfaces, which are then assimilated with resampled IMERG pLP and elevation onto a common hourly 1-km grid.

We train an XGBoost classifier (Chen and Guestrin 2016) on held-out MRoS phase labels and evaluate it in two climatologically distinct ranges, California’s Sierra Nevada and Colorado’s Rocky Mountains, which differ in precipitation pattern, terrain, and observer density. We then benchmark against the standard partitioning methods surveyed by Jennings et al. (2025), using clustered resampling to quantify significance and produce confidence intervals.

2. Study domains and period

Two mountain domains are processed independently through an identical pipeline: the California Sierra Nevada near Lake Tahoe (projected coordinate system in UTM Zone 11N, EPSG:26911) and the Colorado Rocky Mountains (projected coordinate system in UTM Zone 13N, EPSG:32613). Each is bounded by a polygon over mountain terrain, and the study period spans 1 October 2022 to 1 May 2026, excluding June, July and August as the summer non-observation season. All datasets are harmonized to a common 1-km analysis grid whose coordinates and elevations derive from a digital elevation model (DEM). The domains contrast usefully (Figure 1): California has much wider covariation of air temperature with elevation, atmospheric-river-driven storms, and a comparatively balanced observation mix, whereas Colorado’s continental snow climate is strongly snow-dominant with sparser rain.

3. Data

a. Crowdsourced phase observations

MRoS citizen scientists submit timestamped, geolocated reports of precipitation phase (rain, snow, or mixed) during storm events (Arienzo et al. 2021). Volunteers who have opted into storm text alerts, issued when forecast surface temperatures fall in the rain–snow uncertainty range of roughly −2 to 10°C, open a browser-based app during precipitation and record what phase is currently falling. The device supplies geolocation and timestamp automatically. The protocol operationalizes the phase definitions of Section 1: observers report what is falling rather than what has accumulated, classify graupel as snow and freezing rain as rain, exclude hail and true sleet, and reserve the mixed label for genuinely ambiguous or rapidly transitioning precipitation. That last instruction matters, because it makes the mixed label an explicit report of observer uncertainty rather than a claim about a distinct hydrometeor type. Neither temperature nor intensity is entered by the observer,

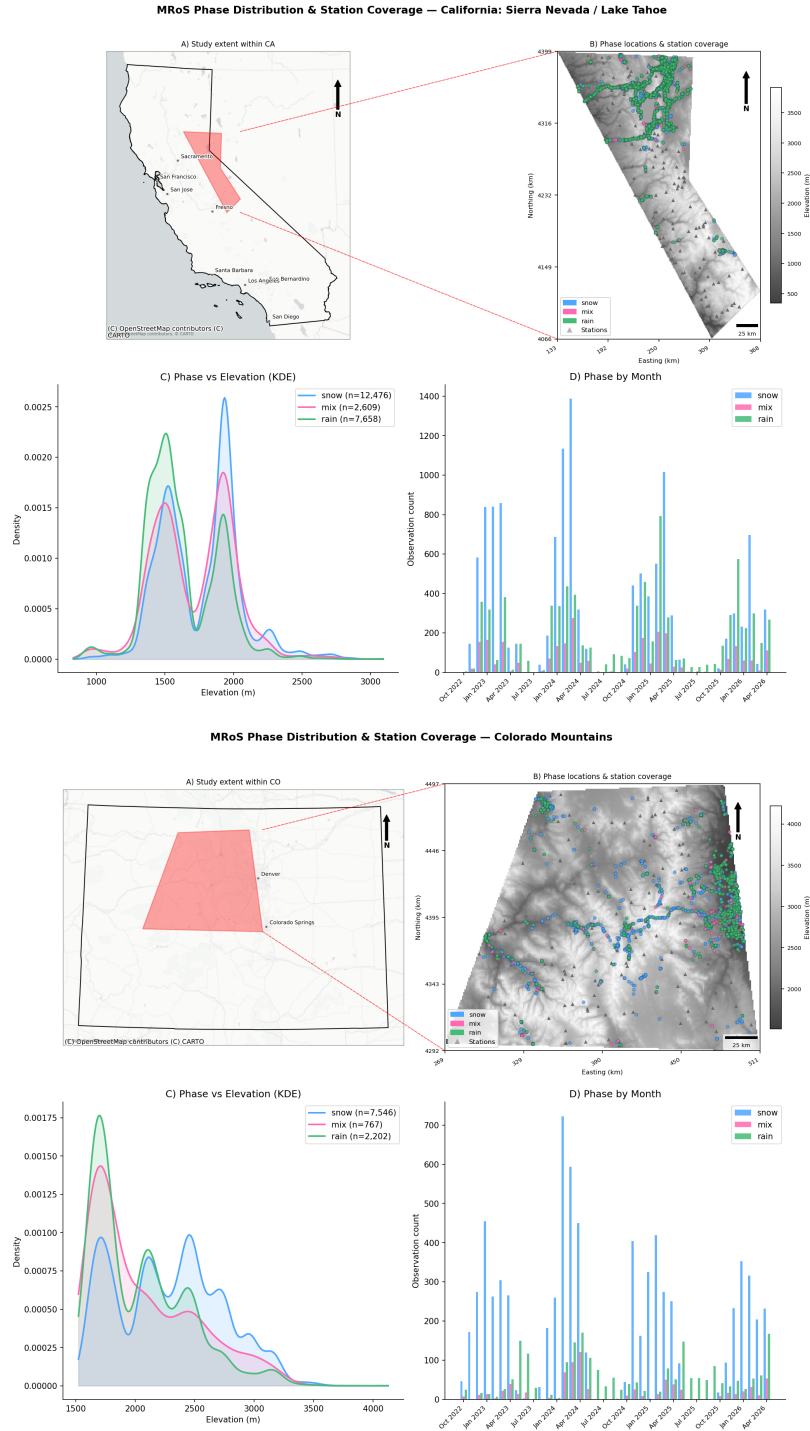


FIG. 1. Study domains and observation distribution for (a, top) California's Sierra Nevada and Lake Tahoe and (b, bottom) Colorado's Rocky Mountains. Within each block: (A) study extent (red) within the state; (B) MRoS reports over shaded relief, colored by reported phase, with assimilated surface stations as grey triangles; (C) kernel density of report elevation by phase; and (D) monthly report counts by phase. Panel C is density-normalized within each phase, so the curves compare shape rather than volume, and phase totals appear in the legend. California's reports are bimodal in elevation, rain-dominated near 1500 m and snow-dominated near 1950 m, whereas Colorado's peak near 1700 m. Two sampling caveats apply throughout: reports cluster along road corridors and near population centers, and monthly counts are participation-driven with near-total summer gaps, so neither panel represents storm climatology.

so each record reduces to a geotagged, timestamped phase label.

Each submission is processed with the `rainOrSnowTools` package (Mountain Rain or Snow Team (Lynker, Desert Research Institute, and University of Nevada, Reno) 2024), which tags it with USGS 3DEP elevation and with modeled surface meteorology from stations within a one-degree radius (requiring at least five stations) after standardized range and consistency checks. Quality assurance follows the procedures established for the network (Arienzo et al. 2021; Jennings et al. 2023): only real-time reports are retained, since ancillary fields are matched to each report’s exact time and location, and observations lacking nearby station support or failing physical plausibility checks are removed. After this, the rain–snow observation set comprises 15,337 observations in California (10,735 train / 2,301 validation / 2,301 test; 64% snow) and 3,492 in Colorado (2,444 / 524 / 524; 82% snow). Observer-reported mixed events constitute a further 12.2% of California test observations (320 of 2,621) and 10.9% of Colorado’s (64 of 588); they are held aside from binary training and used for uncertainty-band evaluation (Section d).

b. Station meteorology

Hourly surface observations are collected from four public networks (HADS, LCD, WCC, and MADIS) via the `rainOrSnowTools` access layer (Mountain Rain or Snow Team (Lynker, Desert Research Institute, and University of Nevada, Reno) 2024). The retained variables are air temperature, dewpoint, and relative humidity, from which wet-bulb temperature is derived during assimilation (Section a). These stations provide the thermodynamic environment that is interpolated to the analysis grid.

c. Satellite probability of liquid precipitation

Half-hourly GPM IMERG probability-of-liquid-precipitation (pLP) fields (Huffman et al. 2020) are retrieved from NASA GES DISC via OPeNDAP, subset server-side to each domain, and merged across the 48 half-hourly timesteps into daily gridded tables. pLP is an independent, satellite-derived estimate of precipitation phase (only rain or snow) from passive microwave and infrared retrievals and later becomes a key model predictor: it carries information about the precipitating column that near-surface measurements cannot typically supply.

d. Reanalysis climate fields

We explored daily PRISM rasters (Daly et al. 2008) as a spatially complete backdrop to the sparse station network, resampling five variables from 800 m to the 1-km grid. PRISM regridded cleanly in space, however its daily

resolution is too coarse for an hourly phase product. Expanding it to hourly required imposing a diurnal temperature curve, and that synthetic sub-daily structure did not improve validation skill over the station-interpolated fields while risking bias near freezing, so we excluded it. Coarser global reanalyses (ERA5, MERRA-2) were similarly rejected because their 30–50-km resolutions cannot resolve the kilometer-scale elevational gradients that control phase here.

e. Terrain

Raw 10-m DEMs from the USGS 3DEP / National Map are clipped to each domain, reprojected to the regional UTM CRS, and resampled to 1 km by average pooling. The resulting DEM defines the analysis grid and supplies the elevation field used for lapse-rate detrending, kriging drift, and as a model predictor.

Table 1 lists the final predictor set, which comprises eight fields once the three MRoS surfaces are counted separately (Section b). Relative humidity is interpolated and used to derive dewpoint and wet-bulb temperature, but is not itself supplied to the model, to avoid redundancy with the other thermodynamic predictors (Section d).

4. Methods

The methodology divides into two pipelines run identically per domain (Figure 2). A data-preparation pipeline (Sections a–b) fuses station meteorology, external gridded products, and MRoS reports, then interpolates them onto the common analysis grid, producing gridded predictor fields and leakage-safe MRoS phase-probability predictors. A modeling pipeline (Sections c–3) trains and calibrates the binary classifier, derives the mixed-phase uncertainty band, and evaluates the result through ablation, benchmarking, and a clustered bootstrap.

a. Data assimilation and quality control

An assimilation stage cleans the station record, derives consistent meteorology, and harmonizes every source to the DEM grid before interpolation. Missing thermodynamic variables are reconstructed from physical relationships. Saturation vapor pressure follows Magnus–Tetens, and dewpoint and relative humidity are interchanged to fill whichever is missing. Wet-bulb temperature, the most physically informative single predictor of phase, is estimated with Stull’s formula (Stull 2011), refined by an iterative psychrometric solution and constrained between dewpoint and air temperature. Where a station reports air temperature without co-located humidity, dewpoint and relative humidity are gap-filled at the station using lapse-rate-adjusted inverse-distance weighting (IDW) from neighboring stations, following the `model.meteo` procedure of `rainOrSnowTools` (Mountain Rain or Snow

TABLE 1. Model predictors and their sources. All fields are harmonized to the hourly 1-km analysis grid.

Predictor	Source	Native resolution
Air temperature	HADS/LCD/WCC/MADIS stations, interpolated	point, hourly
Dewpoint temperature	Stations, interpolated	point, hourly
Wet-bulb temperature	Derived (Stull 2011), interpolated	—
Elevation	USGS 3DEP DEM	10 m
IMERG pLP	GPM IMERG (Huffman et al. 2020)	0.1°, half-hourly
MRoS p_{snow} , p_{mix} , p_{rain}	LOOCV-interpolated MRoS reports	point, hourly

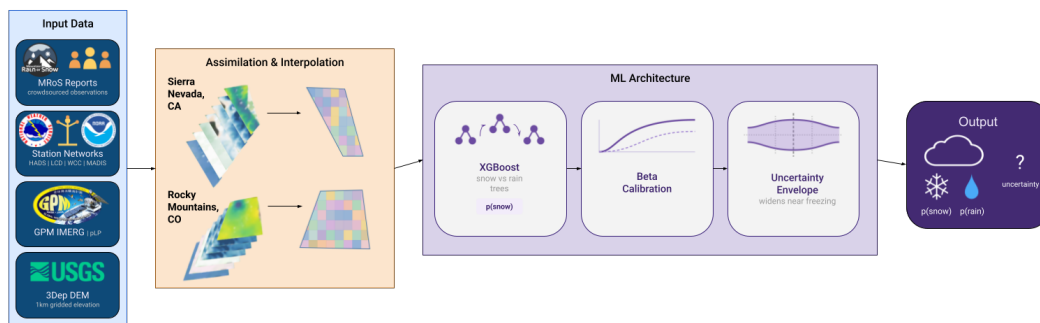


FIG. 2. Data-assimilation and machine-learning architecture. (a) Inputs: crowdsourced MRoS phase reports, surface station networks (HADS, LCD, WCC, MADIS), GPM IMERG probability of liquid precipitation (pLP), and the USGS 3DEP elevation model resampled to the 1-km grid. (b) Station observations and MRoS reports are kriged to that grid per domain and timestamp, producing the predictor stack sampled at each report location. (c) An XGBoost snow-versus-rain classifier is fit separately per domain on eight predictors, its scores are beta-calibrated, and the wet-bulb-conditioned band of Equation 1 is applied. (d) Output is snow, rain, and an uncertainty flag.

Team (Lynker, Desert Research Institute, and University of Nevada, Reno) 2024).

Station observations then pass through quality control: duplicates are dropped, hard physical cutoffs remove impossible readings, a per-station interquartile-range filter removes outliers, spike and stuck-sensor detection nulls implausible runs, and cross-variable checks reconcile temperature with humidity. Survivors are aggregated to one record per station per hour. A dynamic lapse rate is estimated at each timestep from the station temperature–elevation relationship, letting the vertical structure vary hour to hour, and DEM elevation is appended to every point dataset so all sources share an elevation reference. IMERG is brought to the hourly 1-km grid by bilinear resampling, and all retained fields are synchronized on a common UTC axis and written to hourly NetCDF stacks.

b. Spatial interpolation

All point observations are interpolated to the 1-km analysis grid by kriging, the best linear unbiased predictor for a spatial field (Cressie 1993; Goovaerts 1997), with a spherical model fitted to the empirical semivariogram. Because hourly observations are sparse, we pool the empirical semivariogram across time before fitting, which stabilizes the covariance estimate during thinly observed hours.

Each variable is kriged in the form matched to its physical nature. Air and dewpoint temperature carry a strong elevation trend, so they are detrended against station elevation with the hourly lapse rate, interpolated by ordinary kriging on the residuals, and retrended with DEM elevations, a regression-kriging that preserves physical vertical structure (Hengl et al. 2007). Relative humidity is interpolated by universal kriging with DEM elevation as an external drift term (Wackernagel 2003) to reflect terrain dependence, even though it is used only to derive dewpoint and wet-bulb temperature rather than enter the model as a predictor.

Precipitation phase is nominal, so encoding snow, mix, or rain ordinally and interpolating would impose artificial arithmetic (i.e., mix would represent a meaningless average of snow and rain). We therefore convert phase to one-hot indicators, krige each class independently by indicator kriging (Journel 1983), then clip to non-negative values and renormalize so the associated phase probabilities sum to one, yielding hourly surfaces p_{snow} , p_{mix} , p_{rain} . Figure 3 shows the full set at one storm timestamp per domain.

(i) *Leakage prevention.* MRoS observations serve both as training labels and—through the interpolated phase surfaces—as candidate predictors. Reading an observation’s predictor from a full-grid surface built with that same observation would let the label leak into the feature.

To break the dependence we compute the MRoS phase predictors by pointwise spatial leave-one-out cross-validation (LOOCV): each report is removed from its hour, and the remaining reports in that hour are re-kriged by indicator kriging to the held-out location under the same minimum-neighbor requirement. The resulting triplet is clipped, renormalized and stored as the leakage-safe predictor, leaving the phase implied by other observers nearby, which is legitimate spatial context available at inference time rather than the target label itself. Full-grid surfaces built from all observations are retained separately for mapping and gridded prediction, never as training features. This mirrors the leave-one-out logic used for unbiased local estimates in geostatistical cross-validation (Goovaerts 1997).

c. Machine learning model

We frame phase prediction as binary classification and train an XGBoost model (Chen and Guestrin 2016) with a logistic objective on rain–snow observations only. We refer to this eight-predictor configuration as “MRoS-XGB” throughout for simplicity. The output is the probability of snow, with $p(\text{rain}) = 1 - p(\text{snow})$. Gridded predictors are sampled at each MRoS observation location from the hourly NetCDF stacks, giving eight predictors: air, dew-point, and wet-bulb temperature, elevation, IMERG pLP, and the three leakage-safe LOOCV MRoS phase probabilities. Data are split 70/15/15 into train/validation/test, stratified on observed phase class with a fixed random seed, and early stopping is applied on the validation subset (up to 2,000 rounds, patience 50; selected iterations 323 in CA and 513 in CO).

We discarded two alternatives. A three-class model of snow, mix and rain was not pursued because mixed precipitation does not form a separable cluster in feature space (Section d), and spatially blocked splits were not used because blocking left the calibration sets too small to fit stably. The randomized stratified split adopted instead preserves the distribution of meteorological states across subsets, matching the design of Jennings et al. (2025), and defers spatial correlation to the inference stage (Section 3). Its cost is that observations from the same storm may appear in both train and test, so test performance may be mildly optimistic for a novel season.

(i) *Class imbalance.* Both training sets skew toward snow (64% in California, 82% in Colorado). The MRoS network favors higher-elevation, snowier sites, and those sites are also colder, drier, and carry higher LOOCV snow probabilities, so the class imbalance is entangled with the feature space and the standard remedy of frequency-inverse weighting, which balances effective counts rather than the recall asymmetry itself, can over- or under-correct. We therefore tune XGBoost’s `scale_pos_weight` against the quantity of interest, training a probe model across a grid of candidate weights and selecting the value that minimizes

the absolute snow–rain recall gap on the validation set (0.50 for California, 0.15 for the more snow-dominant Colorado). Both fall below one, downweighting the majority snow class, and the resulting recalls balance to within 0.03 and 0.06 respectively (Section a). Because class weighting shifts the raw score distribution, calibration follows it rather than preceding it.

(ii) *Probability calibration.* A classifier can rank cases correctly while reporting probabilities that do not verify, and gradient-boosted trees are particularly prone to this because each boosting round pushes scores toward 0 and 1, an overconfidence the class weighting above amplifies. That matters here because downstream applications need a $p(\text{snow})$ that means what it reports, and because the uncertainty envelope of Section d is an interval in probability space whose placement is meaningful only if the probabilities are.

We therefore fit a post-hoc map from raw score to calibrated probability, and two features of the problem shape its construction. First, near 0°C wet-bulb the thermodynamic signal is weak and true probabilities compress toward 0.5, while away from freezing they approach 0 or 1. One map cannot serve both regimes, so we fit separate calibrators for near-freezing ($|T_w| \leq 2^\circ\text{C}$, consistent with the Sims and Liu 2015 uncertainty range) and away-from-freezing observations, routing each by its wet-bulb temperature.

Second, the choice of calibrator matters more than usual for a well-separated classifier. Isotonic regression, the conventional non-parametric choice, failed diagnostically here: held-out scores fell outside the range seen during fitting, and the out-of-range clipping mapped 62% of test predictions to exactly 0 or 1, which is the signature of a calibrator reintroducing the overconfidence it was meant to remove. We therefore adopt beta calibration (Kull et al. 2017), which is parametric, monotone, and defined on the open unit interval, so it cannot produce degenerate outputs, and whose three parameters let the map be asymmetric where the score distribution is. Calibrators are fitted on final-booster scores over the pooled train and validation rain–snow observations and applied forward to the untouched test set, with mixed observations excluded entirely. Calibration adjusts probability magnitudes only, leaving rank ordering and hard decisions unchanged.

d. Handling mixed phase with an uncertainty envelope

Because rain and snow are more physically defined than mix (Section 1), standard treatments tend to mishandle that asymmetry. Discarding observer-reported mix, as most partitioning studies do, throws away the observations closest to the very transition the model is supposed to resolve. Training mix as a third class instead presumes it is a well-defined category, which it is not: mixed labels are only 11 to 12% of the total, they overlap both pure phases across the same near-freezing temperatures rather than occupying

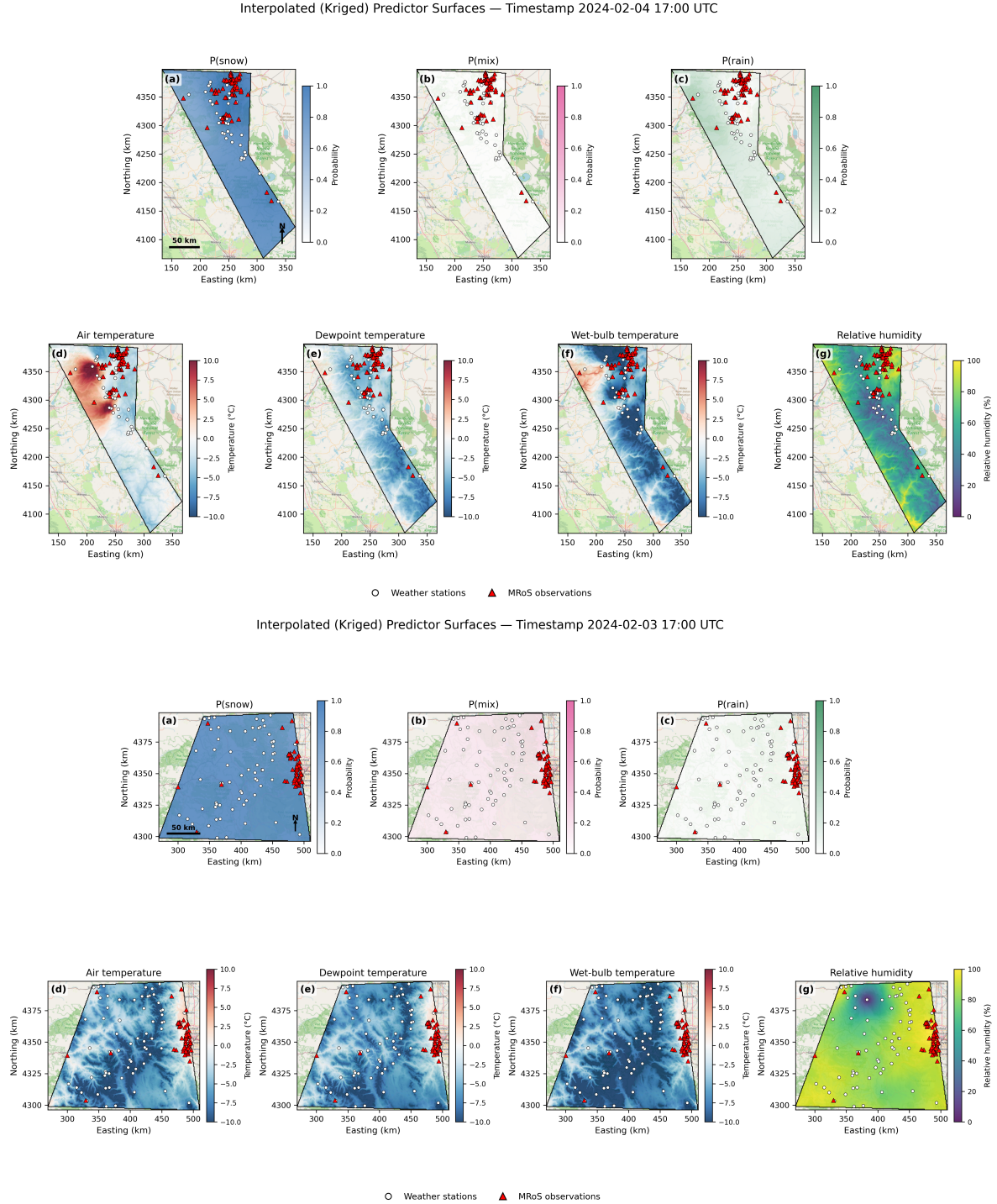


FIG. 3. Kriged predictor surfaces on the 1-km analysis grid for (a, top) California at 2024-02-04 17:00 UTC and (b, bottom) Colorado at 2024-02-03 17:00 UTC, two independent storm timestamps. Within each block, panels (a)–(c) are the kriged MROs phase probabilities and panels (d)–(g) are kriged air, dewpoint and wet-bulb temperature and relative humidity. Open circles mark contributing stations, red triangles MROs reports at that timestamp. The fields are smooth by construction, representing mesoscale structure rather than local microclimate, and predictor quality degrades away from the observation clusters, most visibly in the sparsely instrumented southern Sierra. Note that these are the full-data surfaces; the model predictor is the noisier leave-one-out version, which excludes the training report at its own location.

a distinct region of feature space, and observers can apply the label inconsistently. A trained third class therefore learns mostly label noise while degrading the rain–snow boundary, which is what our first multiclass iteration found (Section c).

We take a third route: keep the supervised problem binary, and let mix be represented by a region of uncertainty in the model’s own output rather than by a class label. Where that region should sit is not obvious, since a fixed threshold pair such as 0.30/0.70 would be physically unjustified when the meaning of an intermediate score depends on thermodynamic context. Near 0°C wet-bulb, where snow and rain routinely coexist in the same column, $p(\text{snow}) = 0.65$ is genuinely ambiguous; at $T_w = -5^\circ\text{C}$ the same score would instead signal a real anomaly. We therefore let the threshold width vary continuously with wet-bulb temperature through a Gaussian half-band $h(T_w)$ applied post-hoc to the calibrated probabilities:

$$\begin{aligned} h(T_w) &= b + a \exp\left(-\frac{T_w^2}{2\sigma^2}\right), \\ \text{rain threshold} &= 0.5 - h(T_w), \\ \text{snow threshold} &= 0.5 + h(T_w), \end{aligned} \quad (1)$$

The band is centered at $T_w = 0^\circ\text{C}$ and decays as a Gaussian of width σ , with b setting the minimum half-width far from freezing and a the additional widening at 0°C (Figure 8). Any observation whose calibrated $p(\text{snow})$ falls inside the band is flagged as uncertain rather than assigned a phase. We call this an *uncertainty envelope* to emphasize that it does not predict a mixed class but delineates, in probability space, the zone within which the binary model loses confident discrimination and thus abstains from making a distinct call. Observer-reported mix events then serve as an independent check on whether that zone lands where real mixed precipitation occurs.

Two metrics follow from this construction, defined in Table 2: the *flag rate* or abstain rate, a ground-truth independent property of model behavior, and the *mix-capture rate*, conditional on observer labels. They vary independently, since a model can flag frequently and still miss real mix if the envelope is centered wrongly, or flag rarely and capture most of it if the envelope is well placed.

The three band parameters are optimized on the validation set, which for this purpose alone includes the observed mixed events. We optimize a composite of mix capture and confident rain–snow accuracy, subject to a minimum rain–snow coverage constraint, with σ bounded to physically reasonable transition-zone widths (Sims and Liu 2015; Jennings et al. 2025). The fitted parameters are identical in both domains ($b = 0.20$, $a = 0.15$, $\sigma = 2.0^\circ\text{C}$), placing the thresholds at $p < 0.15$ for rain and $p > 0.85$ for snow at 0°C wet-bulb and relaxing them to 0.30 and 0.70 far from freezing.

e. Evaluation framework

The framework is evaluated in three ways. A feature ablation (Section 1) retrains the model with inputs withheld to establish which sources the skill depends on, a benchmark comparison (Section 2) scores it against operational partitioning methods overall and near freezing, and the uncertainty envelope is assessed against observer-reported mixed events (Section d). Every result is reported with a confidence interval, and every comparison as a difference with an interval of its own (Section 3).

Every metric is additionally computed on a *near-freezing subset*, defined throughout as $|T_w| \leq 2^\circ\text{C}$ after the Sims and Liu (2015) uncertainty range. This is an observation-level selection in wet-bulb temperature and should be distinguished from the 0°C – 4°C range of 50% rain–snow air-temperature thresholds reported across sites by Jennings et al. (2018), which some figures mark for reference. The two describe much the same physical regime, since wet-bulb temperature sits below air temperature in the unsaturated conditions typical of these storms, but the former selects individual observations while the latter summarizes variation between locations.

(i) *Metrics.* A suite of complementary metrics reflects model performance better than any individual metric: whether phase is correct on rain as well as on the dominant snow class, whether output probabilities can be trusted at face value, and whether model abstention falls on the right mix cases. Table 2 accordingly groups the metrics by what they measure. Accuracy compared against macro F1 diagnoses class-balance effects, while phase bias compared against accuracy separates aggregate from event-level correctness (Section d). Because a committed call discards the confidence behind it, a second group scores the probabilities directly. A third scores the mix-related flag rate, which has no counterpart in the rain–snow benchmarks and is the only group computed over mixed-phase observations.

1) ABLATION DESIGN

Deciding which inputs an operational pipeline should maintain requires knowing not only which features a model uses but which it needs. We use SHapley Additive exPlanations (SHAP) (Lundberg et al. 2020) to identify which features the fitted model leans on and to select which ablation configurations are worth running, then treat the retrainings as the causal evidence, since attribution alone cannot establish what would happen to skill if an input were unavailable.

Seven configurations are evaluated per domain, listed with their retained predictors in Table 4, in three groups: the full eight-predictor configuration, leave-one-out removals isolating each assimilated source in turn, and reduced sets testing collective value (a thermodynamics-only configuration and two reduced cores). All share identical

TABLE 2. Evaluation metrics. Confusion-matrix quantities take snow as the positive class: true positive (TP) and true negative (TN) are correctly called snow and rain, false positive (FP) is rain called snow, false negative (FN) is snow called rain, and N is the number of rain and snow observations. Probability metrics use the calibrated $p(\text{snow})$, written p_i for observation i with outcome $y_i \in \{0, 1\}$ (1 = snow). Envelope metrics use the uncertainty band of Equation 1 and are the only ones computed over all N_{all} observations, mix included, where M is the number of observer-reported mixed events, M_{in} those inside the band, and F all predictions inside it. Every metric is also computed on the near-freezing subset ($|T_w| \leq 2^\circ\text{C}$).

Metric	Description	Definition	Limitation
<i>Confusion-matrix metrics</i>			
Accuracy	How often the predicted phase matches the observed phase.	$(\text{TP} + \text{TN})/N$	Dominated by the majority class.
Snow recall	The share of observed snow the model identifies.	$\text{TP}/(\text{TP} + \text{FN})$	Ignores false alarms.
Rain recall	The share of observed rain the model identifies.	$\text{TN}/(\text{TN} + \text{FP})$	Noisy where rain is rare.
Macro F_1 score	Precision and recall combined for each phase, then averaged so both phases count equally however often they occur.	Mean of F_1^{snow} and F_1^{rain} , where $F_1^{\text{snow}} = 2\text{TP}/(2\text{TP} + \text{FP} + \text{FN})$	Uses the committed call only, not the probability behind it.
Critical success index (CSI)	The share of events either observed or predicted that the model called correctly, computed once with snow as the event class and once with rain.	$\text{TP}/(\text{TP} + \text{FP} + \text{FN})$, per phase	Excludes correct negatives, so it depends on how often the event occurs and is not comparable across temperature bins.
Phase bias (%)	Whether a phase is predicted more or less often than it was observed.	$100(n_{\text{pred}}/n_{\text{obs}} - 1)$, per phase	Over- and under-calls cancel.
<i>Predicted-probability metrics</i>			
Area under the ROC curve (AUROC)	How well the probabilities rank snow above rain, at any threshold.	$\Pr(p_i > p_j)$ for a random observed-snow i and observed-rain j	Unaffected by whether the probabilities are calibrated.
Brier score	How far the stated probabilities lie from the observed outcome.	$N^{-1} \sum_i (p_i - y_i)^2$	Mixes calibration with confidence.
<i>Uncertainty-envelope metrics</i>			
Mix-capture rate	The share of observer-reported mixed events falling inside the envelope.	M_{in}/M	Rises automatically as the band widens.
Flag rate	How often the model declines to commit to a phase (abstain).	F/N_{all}	Says nothing about whether declining was correct.

splits and the same sampled predictor matrix, so differences are attributable only to the feature configuration, and all are scored on the metrics of Table 2 with near-freezing values tracked separately.

2) BENCHMARK METHODS

We compare against the standard precipitation-phase partitioning methods surveyed by Jennings et al. (2025), implemented as deterministic functions of the same gridded predictor fields: static air-temperature thresholds ($T_a = 1.0$ and 1.5°C), dewpoint thresholds ($T_d = 0.0$ and 0.5°C), and wet-bulb thresholds ($T_w = 0.0, 0.5$, and 1.0°C). Two logistic methods are also included: the binary logistic re-

gression of Jennings et al. (2018),

$$p(\text{snow}) = \frac{1}{1 + \exp(-10.04 + 1.41 T_a + 0.09 \text{RH})}, \quad (2)$$

and the same logistic form re-fitted to each domain’s training split, which isolates the benefit of local recalibration from model form. All methods are evaluated on the identical held-out test split. Following Jennings et al. (2025), the primary comparison is binary (observer-reported mixed excluded), scored with accuracy, macro F1, and per-phase bias, reported overall and by temperature bin, with the near-freezing regime reported separately. The threshold and logistic benchmarks are strictly binary and cannot predict mixed events, so mixed-phase skill is evaluated separately from the binary comparison.

3) CONFIDENCE INTERVALS AND SIGNIFICANCE

Because a metric computed on a single test set is only one draw from a sampling process, we bootstrap the saved test-set predictions (Efron and Tibshirani 1993), resampling with replacement 5,000 times and recomputing every metric on each copy. Intervals span the 2.5th to 97.5th percentile of the resulting values. Point estimates are computed once on the full test set rather than as medians over resamples, so intervals need not be symmetric about them and are markedly asymmetric on small subsets. All are conditional on the trained model and the realized split, and quantify test-set sampling uncertainty only, not retraining variability.

Two departures from the standard procedure are needed. First, MRoS observations are not independent, since many reports duplicate one another within a storm-day, so resampling individual rows would credit the test set with more evidence than it holds and give intervals that are too narrow. We therefore resample intact clusters, defined as the observations sharing a 30-km spatial block and a calendar day. Interval widths are essentially identical across 15-, 30- and 60-km blocks, so the imposed geometry does not drive the conclusions, although clustering itself widens intervals by 13 to 28%. Second, methods are compared inside each resample rather than by asking whether two separately computed intervals overlap, since the scores being compared, whether two methods or two feature configurations, rise and fall together from one resample to the next. Scoring all methods on identical resampled rows and recording the difference cancels that shared variability, for the same reason a paired test outperforms a two-sample test on the same data. One-sided p -values are the fraction of resamples in which the benchmark matched or exceeded the model.

5. Results

Reported skill values are point estimates with 95% confidence intervals, and comparisons between methods or feature sets are differences computed on identical resampled data, each carrying an interval of its own (Section 3). Unless stated otherwise, all values are held-out test-set values; the few validation-set quantities are labeled as such. Primary performance results for both domains are collected in Table 3, per-method benchmark results in Table 5, and per-configuration ablation results in Table 4.

a. Overall model skill

MRoS-XGB performs comparably in the two domains despite their contrasting snow climates, with a test AUROC of 0.965 in both and an accuracy of 91.2% in California and 91.4% in Colorado (Table 3; Figure A1 gives the corresponding confusion matrices). Colorado’s intervals are

roughly three times wider than those of California’s, reflecting its smaller test set (524 rain–snow observations in Colorado against 2,301 in California).

Class weighting balances the two recalls as intended, to within 0.03 in California and 0.06 in Colorado despite snow shares of 64% and 82% (Table 3), and it does so at negligible cost in aggregate skill, since macro F1 changes by less than 0.01 in both domains. Correcting the recall asymmetry therefore does not trade against overall performance.

What remains of the imbalance appears in phase bias, the percentage by which each phase was predicted more or less often than it was observed. Both California intervals contain zero, so no systematic over- or under-prediction is detectable there, whereas Colorado predicts snow 4.4% less often than observed and rain 19.8% more often, neither interval containing zero. Its rain interval is wide because the test set holds only 96 rain observations, but the direction is resolved: the strong snow downweighting imposed by the optimized class weight (`scale_pos_weight` = 0.15) balances recall by over-correcting the predicted totals toward rain.

Beta calibration pulls the raw scores toward the reliability diagonal in both domains (Figure A2), giving calibrated Brier scores of 0.070 in California and 0.058 in Colorado. What residual miscalibration remains concentrates at the extremes of $p(\text{snow})$, where the score distribution is strongly bimodal and the bins are therefore heavily populated.

b. Skill in the near-freezing regime

Accuracy falls near freezing in both domains, as expected. On the near-freezing subset ($|T_w| \leq 2^\circ\text{C}$; $n = 782$ in California, $n = 179$ in Colorado), accuracy drops 5.6 and 8.7 percentage points below each domain’s overall figure. AUROC falls much less, from 0.965 to 0.922 and 0.920 (Table 3), so the model continues to rank snow above rain reliably even where its committed calls are least accurate. This subset is defined by wet-bulb temperature and is not the same population as the observations the model declines to call, which is defined in probability space (Section e); the two overlap heavily but are scored separately throughout.

Resolving skill into 1°C wet-bulb bins shows that the two domains lose accuracy in structurally different ways (Figure 4). California’s per-phase F1 is non-monotonic in T_w , with a minimum near 1.5°C , whereas Colorado’s degrades smoothly as T_w warms, and the breakdown against air temperature (Figure A3) reproduces both structures. California’s wider covariation of temperature with elevation and its bimodal distribution of observation elevations (Figure 1) would plausibly produce several distinct near-freezing regimes where Colorado’s continental climate produces one, although the present sample cannot separate those contributions.

TABLE 3. Held-out test-set point estimates with 95% cluster-bootstrap confidence intervals (5,000 resamples; 30-km $\times$ calendar-day clusters; percentile intervals). Point estimates are computed once on the full test set and are not medians over resamples, so intervals need not be symmetric about them. Δ rows are differences against the best benchmark in each domain ($T_d = 0.0^\circ\text{C}$ in CA; $T_d = 0.5^\circ\text{C}$ in CO). Phase bias is the percentage by which each phase was predicted more (+) or less (−) often than observed. All intervals are conditional on the trained model and realized split.

Quantity	California	Colorado
Accuracy (%)	91.2 (89.9–92.5)	91.4 (88.4–94.1)
Near-freezing accuracy (%)	85.6 (82.9–88.3)	82.7 (76.4–88.3)
AUROC	0.965 (0.957–0.972)	0.965 (0.942–0.982)
Near-freezing AUROC	0.922 (0.903–0.941)	0.920 (0.865–0.962)
Macro F1	0.905 (0.891–0.920)	0.867 (0.821–0.906)
Near-freezing macro F1	0.855 (0.828–0.882)	0.820 (0.753–0.877)
Snow recall	0.923 (0.908–0.938)	0.925 (0.894–0.953)
Rain recall	0.893 (0.868–0.916)	0.865 (0.789–0.930)
Snow phase bias (%)	−1.6 (−3.5+0.4)	−4.4 (−7.8−1.3)
Rain phase bias (%)	+2.9 (−0.7+6.4)	+19.8 (+5.6+37.8)
Mix-capture rate	0.30 (0.25–0.36)	0.33 (0.21–0.45)
Δ accuracy vs. best benchmark (pts)	+11.1 (9.0–13.4)	+3.6 (0.2–6.9)
Δ near-freezing accuracy vs. best (pts)	+16.5 (12.2–21.1)	+8.4 (−0.6–16.9)

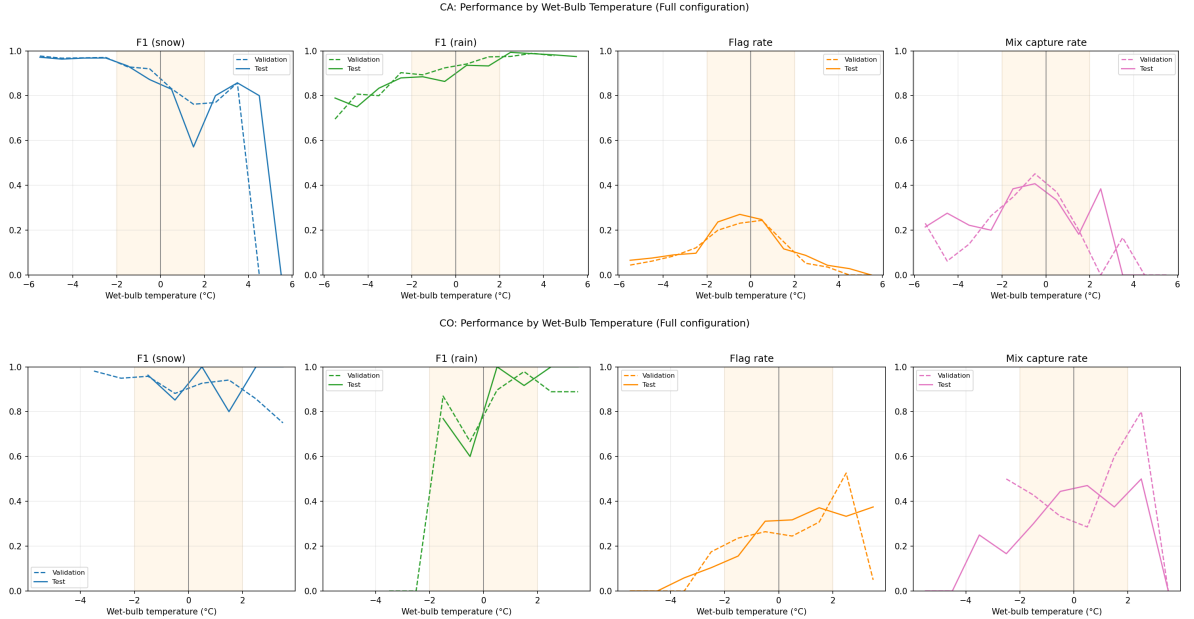


FIG. 4. Per-phase F1, flag rate (a.k.a., abstain rate), and mix-capture rate in 1°C wet-bulb bins for California (top) and Colorado (bottom), validation (dashed) and test (solid). Shading marks $|T_w| \leq 1^\circ\text{C}$; bins span -6°C to 6°C and are plotted at midpoints. Skill is lowest, and the uncertainty envelope most valuable, in the near-freezing transition zone. The F1 panels count only observations the model committed to, so they should be read against the flag-rate panel: a high F1 brought by frequent abstention is not the same result as a high F1 at full coverage. Line breaks mark undefined or under-sampled statistics, arising where a bin’s committed sample holds one phase or fewer than 10 observations. A plotted zero instead reflects the model’s failure to recover a phase that was present.

c. Input-dependent skill

Our first research question asks whether the assimilated predictors contribute skill beyond thermodynamics. We answer it in two steps, asking first which predictors the fitted model uses and then which it needs.

(i) *Feature attribution.* SHAP attributions show that the two domains lean on different predictors (Figure A4).

California is led by the leakage-safe MRoS surfaces and elevation, with the raw temperatures contributing little, whereas Colorado is led by IMERG pLP and air temperature. Comparing attribution on the near-freezing subset with the clear-phase subset, elevation and the MRoS surfaces gain importance in both domains, and in Colorado the three MRoS surfaces gain the most of any predictor (ratios of 1.6–1.9). The interpolated MRoS mix surface is among

them in both domains, which is worth distinguishing carefully: observer-reported mix is excluded from the training labels but enters the model as a predictor, so mixed reports carry usable signal even though they are never a target, a use prior work has not made of them.

Some inputs are correlated, which limits what attribution can establish. Air and wet-bulb temperature correlate at 0.81 in California and 0.97 in Colorado, the MRoS snow and rain surfaces at -0.92 and -0.89 , and IMERG pLP with the MRoS snow surface at -0.75 in California. SHAP credit assigned to one member of a correlated group is therefore not a statement about how much unique information that predictor supplies, and the ablation addresses this directly. We refer throughout to *configurations* rather than to separate models, since all seven share an identical training procedure, split, and predictor matrix and differ only in which predictors they retain; the full eight-predictor configuration is MRoS-XGB itself. Table 4 gives each configuration’s absolute skill and Figure A5 the paired difference from the full configuration, which is the skill that configuration would forfeit without the withheld input.

In California the MRoS surfaces are the most damaging removal, worth 9.6 percentage points of near-freezing accuracy (6.5–12.7), and dropping all five non-thermodynamic predictors costs 15.7 points (12.2–19.4); both removals are clear of zero on every metric, and no other removal resolves on more than one (Table 4, Figure A5). Three further findings emerge only from the paired differences. First, the three temperature variables are largely interchangeable on the overall metrics, but that interchangeability breaks down at the phase boundary, where wet-bulb temperature is the only one whose contribution the data resolve (1.7 near-freezing points, 0.4–3.0). Second, elevation contributes measurably on every metric, costing 1.7 points of overall accuracy (0.8–2.5) when removed. Third, IMERG pLP contributes specifically near freezing, worth 1.9 points (0.5–3.3) there, while its effect on the overall metrics stays within noise.

Colorado’s smaller test set distinguishes fewer effects, and most of its difference intervals span zero. Three still resolve: dropping all five non-thermodynamic predictors costs 11.2 near-freezing points (3.1–19.5), the MRoS surfaces register on overall AUROC (0.017, 0.004–0.033), and pLP registers on near-freezing AUROC (0.017, 0.001–0.036), the same near-freezing role it plays in California. Near-freezing AUROC summarizes the pattern in both domains, rising under the full configuration from the thermodynamics-only values of 0.796 (0.761–0.830) in California and 0.791 (0.715–0.862) in Colorado to 0.922 and 0.920, where gains of 0.126 (0.096–0.158) and 0.129 (0.066–0.196) persist. Two further configurations test whether a shorter predictor list would serve, and a reduced core of wet-bulb temperature, elevation, pLP and the three MRoS surfaces matches the full configuration on every overall metric in both domains while giving up

2.4 near-freezing accuracy points in California (0.8–4.2). A shortened list is therefore adequate for overall skill but costs measurably near freezing.

Resolving both the ablation and the benchmark comparison for the near-freezing subset separately answers the second research question. In California, removing the MRoS surfaces costs 0.097 of near-freezing macro F1 against 0.037 away from freezing, and removing all non-thermodynamic predictors costs 0.162 against 0.059. The assimilated information therefore contributes in both regimes and contributes about 2.7 times as much at the phase boundary.

Resolving the MRoS ablation against temperature (Figures 5 and A6) shows where that loss is incurred. In California the removal costs recall rather than precision, and it does so for whichever phase is locally in the minority: snow recall halves in the warmest transition bins while snow precision is nearly unchanged, and rain recall falls by comparable margins on the cold side. Without the crowd-sourced surfaces the model therefore defaults toward the phase its thermodynamic setting implies and misses the one that is locally unexpected. The same comparison against air temperature reproduces the rain-side loss but weakens the snow-side loss considerably, consistent with wet-bulb temperature being the sharper discriminator of phase.

d. Benchmarking

MRoS-XGB outperforms all nine benchmarks in both domains (Table 5, shown as ranked bars in Figure A7), and the margin widens near freezing (Figure 6). In California its accuracy is 91.2% against 80.1% for the best benchmark, a dewpoint threshold at $T_d = 0.0^\circ\text{C}$, and 85.6% against 69.1% near freezing. In Colorado it is 91.4% against 87.8% for $T_d = 0.5^\circ\text{C}$, and 82.7% against 74.3% near freezing. The benchmark ordering reproduces Jennings et al. (2025), with air-temperature-only thresholds worst and humidity-aware thresholds intermediate.

We compute a paired model-minus-benchmark difference for each of the nine benchmarks on each of four metrics, 36 differences per domain; Figure 7 plots the accuracy subset. In California all 36 are positive and clear of zero, with no resample out of 5,000 favoring a benchmark ($p < 0.0002$, the smallest value 5,000 resamples can resolve). Against the best benchmark MRoS-XGB gains 11.1 points of accuracy overall (9.0–13.4) and 16.5 near freezing (12.2–21.1), so even at the interval floors it is at least nine points better overall and twelve better near freezing than the strongest method in the suite.

Colorado’s margins are smaller, and there the choice of metric matters. Thirty-five of its 36 differences clear zero, the overall accuracy gain against $T_d = 0.5^\circ\text{C}$ being 3.6 points (0.2–6.9, $p = 0.021$). The near-freezing accuracy gain against the same benchmark, 8.4 points (-0.6

TABLE 4. Ablation study: held-out test performance by feature configuration, with 95% cluster-bootstrap confidence intervals. The predictors retained by each configuration are listed in the third column, using T_a , T_d and T_w for air, dewpoint and wet-bulb temperature, E for elevation, P for the IMERG probability of liquid precipitation, and M for the three leakage-safe MRoS phase surfaces, which count as three predictors; n is the resulting predictor count. All configurations share kriged predictors, beta calibration, and identical splits, so differences are attributable to the feature set alone. Decimal metrics omit the leading zero and NF denotes the near-freezing subset ($|T_w| \leq 2^\circ\text{C}$). Note that these absolute intervals overlap between configurations even where the paired difference is clear of zero, so Figure A5, not this table, is the basis for every significance claim in the text.

Domain	Configuration	Predictors retained	n	ROC AUC	Macro F1	NF accuracy (%)	NF ROC AUC
CA	Full configuration (MRoS-XGB)	T_a, T_d, T_w, E, P, M	8	.965 (.957–.973)	.905 (.891–.919)	85.6 (82.9–88.3)	.922 (.902–.941)
CA	No MRoS surfaces	T_a, T_d, T_w, E, P	5	.936 (.925–.947)	.850 (.832–.867)	76.0 (72.8–79.1)	.861 (.835–.888)
CA	No IMERG pLP	T_a, T_d, T_w, E, M	7	.964 (.956–.972)	.899 (.885–.913)	83.6 (80.8–86.3)	.916 (.895–.936)
CA	No elevation	T_a, T_d, T_w, P, M	7	.956 (.947–.964)	.887 (.873–.902)	82.9 (80.1–85.7)	.905 (.882–.926)
CA	Thermodynamics only	T_a, T_d, T_w	3	.907 (.892–.921)	.817 (.797–.836)	69.8 (66.3–73.3)	.796 (.761–.830)
CA	Reduced core	T_w, E, P, M	6	.966 (.958–.973)	.900 (.885–.914)	83.1 (80.2–85.9)	.922 (.902–.941)
CA	Reduced core, no pLP	T_w, E, M	5	.963 (.955–.971)	.895 (.880–.909)	82.0 (79.0–84.8)	.917 (.895–.937)
CO	Full configuration (MRoS-XGB)	T_a, T_d, T_w, E, P, M	8	.965 (.941–.982)	.867 (.820–.906)	82.7 (76.4–88.3)	.920 (.864–.962)
CO	No MRoS surfaces	T_a, T_d, T_w, E, P	5	.948 (.916–.973)	.863 (.820–.903)	84.4 (78.5–89.7)	.896 (.836–.946)
CO	No IMERG pLP	T_a, T_d, T_w, E, M	7	.959 (.935–.978)	.871 (.825–.910)	82.7 (76.5–88.3)	.903 (.841–.951)
CO	No elevation	T_a, T_d, T_w, P, M	7	.959 (.936–.977)	.862 (.818–.901)	81.6 (75.7–87.0)	.912 (.857–.956)
CO	Thermodynamics only	T_a, T_d, T_w	3	.925 (.897–.951)	.798 (.746–.846)	71.5 (63.8–78.6)	.791 (.715–.862)
CO	Reduced core	T_w, E, P, M	6	.963 (.940–.981)	.862 (.816–.903)	81.6 (75.6–87.4)	.924 (.873–.965)
CO	Reduced core, no pLP	T_w, E, M	5	.955 (.930–.974)	.853 (.803–.896)	80.4 (73.7–86.6)	.901 (.842–.948)

to +16.9), is the single difference that does not, although 96% of resamples still favor MRoS-XGB ($p = 0.037$).

On macro F1 every Colorado difference clears zero ($p \leq 0.0002$). Macro F1 weights rain equally and therefore exposes the collapse in benchmark rain recall that accuracy conceals where 82% of observations are snow. The critical success index reproduces this asymmetry independently of the class balance (Figure A8), since every benchmark’s rain-class CSI falls toward zero on the cold side of its own threshold while MRoS-XGB retains skill on both event classes across the full temperature range. Against $T_d = 0.5^\circ\text{C}$, whose rain recall is 0.531 against MRoS-XGB’s 0.865 (Table 5), the macro F1 gain is 0.096 (0.042–0.152). The two domains therefore support the same conclusion at different strengths. The advantage is unambiguous where the observer network is dense. Where it is sparse it remains unambiguous on the balanced metric but is marginal on the metric the majority class dominates.

(i) *Phase bias.* Phase bias separates the benchmarks more sharply than accuracy does. Near freezing in California, the $T_a = 2.0^\circ\text{C}$ and $T_w = 2.0^\circ\text{C}$ thresholds are set at the same nominal value yet fail by comparable margins in opposite directions, the former calling snow 76% less often than it occurred and the latter 74% more often (Table 5). Wet-bulb temperature sits well below air temperature in the dry, unsaturated conditions typical of these storms, so the wet-bulb cutoff admits far more observations to the snow class. Because these errors partly cancel within a domain, accuracy alone conceals them.

Only two California methods predict each phase about as often as it occurred: MRoS-XGB and the Jennings

et al. (2018) logistic applied without refitting. The transferred logistic reproduces the observed totals but is correct on only 72% of individual events, whereas MRoS-XGB reproduces them at 91% accuracy. Predicting the right amount of snow is not the same as predicting it on the right days, and the latter is useful for forcing a hydrologic model. Figure A9 resolves these biases against air temperature for the full benchmark suite, showing that each threshold’s error reverses sign across its own threshold, so a method can appear accurate in a bin while being badly biased in both phases within it.

e. Uncertainty envelope and mixed precipitation

The third research question concerns the observations the model declines to call. Neither accuracy nor the benchmark suite applies here, since no partitioning method in the comparison considers mixed phase or abstains from a call. Because the envelope is widest at $T_w = 0^\circ\text{C}$ by construction (Equation 1), the flag rate peaking there (Figure 4) reflects the design rather than the data, and the test is instead whether observer-reported mixed events, withheld from training, fall inside it (Figures 8 and A10).

Overall, the envelope abstains near the rate at which mixed precipitation was reported; it flags 12.9% of California test observations against an observed mix fraction of 12.2%, and 14.0% of Colorado’s against 10.9%. The volume of abstention is thus aligned without having been fitted to that quantity.

Capture of individual events is lower, at 30% (95% CI 25–36%) of held-out mixed events in California and 33% (21–45%) in Colorado. The misses are not near-misses,

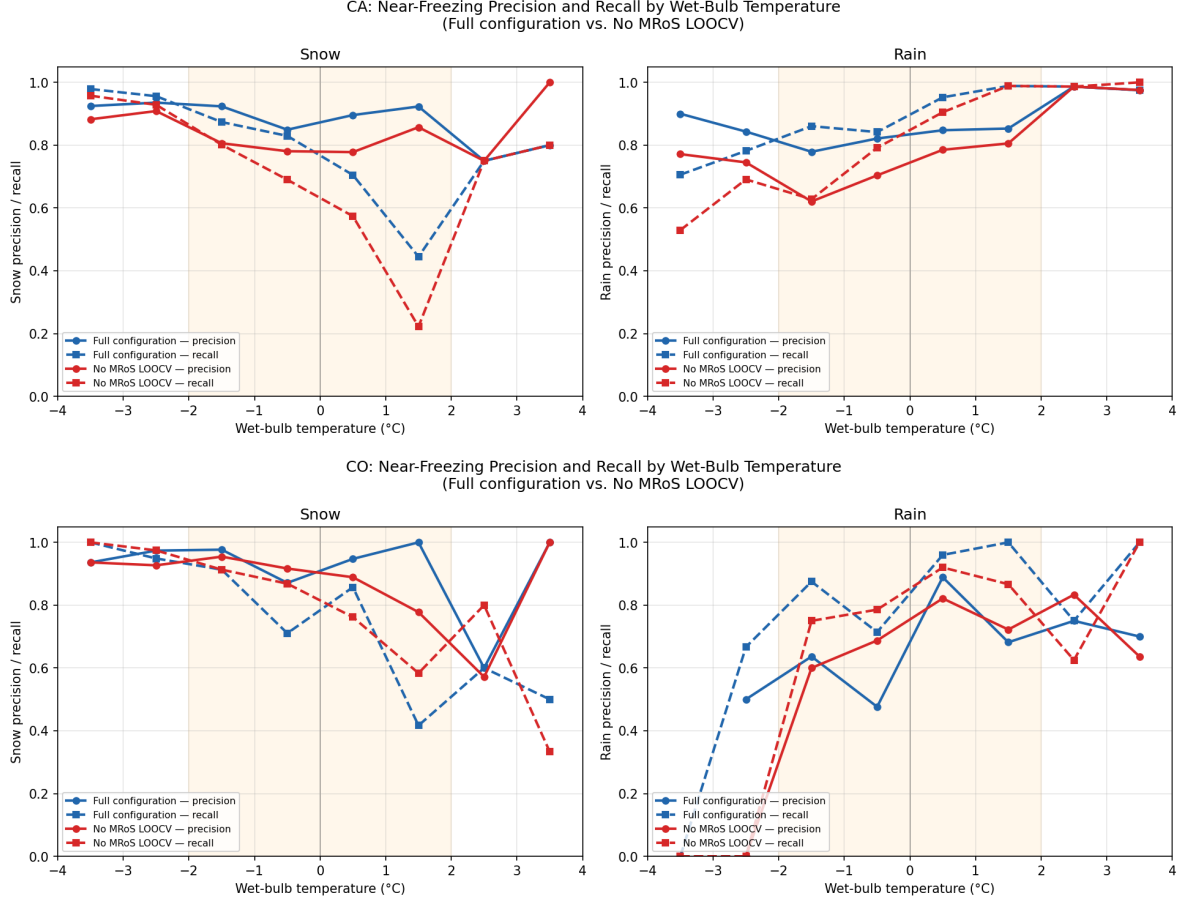


Fig. 5. What the crowdsourced surfaces contribute at the phase boundary: near-freezing precision and recall against wet-bulb temperature for the full configuration (blue) and the same configuration with the MRoS surfaces removed (red), for California (top) and Colorado (bottom), snow (left) and rain (right). Held-out test split, scored at the raw 0.5 threshold with no uncertainty band applied, so the comparison isolates the predictor contribution from the abstention rule; observer-reported mix is excluded and shading marks $|T_w| \leq 2^\circ\text{C}$. Colorado’s curves cross repeatedly rather than separating. Its individual bins rest on tens of observations, so those crossings sit well inside sampling noise and the interval-based ablation of Table 4 remains the appropriate test there. Figure A6 repeats the comparison against air temperature.

since nearly all carry confident probabilities ($p > 0.8$ or $p < 0.2$), so a wider band would recover few of them. The domains also differ in where captured events fall (Figure 8): California’s span a wide range of temperature and probability, whereas Colorado’s cluster near $p \approx 0.5$ – 0.65 and $T_w \in [-1, +2]^\circ\text{C}$.

Band width trades capture against coverage, since a wider band catches more mixed events but declines to call more of the rain and snow the model would have classified correctly. The operating point is selected on the validation set, where it captures 36% of California’s mixed events, commits to 90% of its rain and snow, and is correct on 94.4% of those calls; that figure sits above the 91.2% headline because it counts only committed observations. Narrowing the far-from-freezing half-width from 0.20 to 0.10 raises coverage from 90% to 94% as capture falls from 0.36 to 0.20, with Colorado following the same curve at

uniformly lower capture. Reporting the curve rather than the single point lets users choose their own balance.

6. Discussion

a. A ceiling set by observation type

Jennings et al. (2025) found that machine learning barely improves on threshold rules given near-surface meteorology, and our thermodynamics-only configuration reproduces that ceiling. MRoS-XGB clears it, and because the ablations locate the difference in the assimilated information rather than in model form, the ceiling is better understood as a property of the observations than of the classifier. A limit imposed by model form would erode as architectures improve, whereas one imposed by observation type will not, since every such model is fitted to the same near-surface thermodynamics that cannot resolve

TABLE 5. Held-out test performance of the model and the nine benchmark partitioning methods, with 95% cluster-bootstrap confidence intervals (5,000 resamples; 30-km $\times$ calendar-day clusters). Methods are ordered by overall accuracy within each domain. NF = near-freezing subset ($|T_w| \leq 2^\circ\text{C}$; $n = 782$ in CA, $n = 179$ in CO). Point estimates are computed once on the full test set and are not medians over resamples, so intervals need not be symmetric about them. Per-phase recall and bias for the model are given in Table 3, and the paired model-minus-benchmark differences underlying the significance claims of Section d appear in Figure 7.

Domain	Method	Accuracy (%)	Macro F1	NF accuracy (%)	NF macro F1
CA	MRoS-XGB	91.2 (89.9–92.5)	.905 (.891–.920)	85.6 (82.9–88.3)	.855 (.828–.882)
CA	$T_d = 0.0^\circ\text{C}$	80.1 (77.9–82.2)	.781 (.757–.802)	69.1 (65.2–72.8)	.689 (.651–.726)
CA	$T_d = 0.5^\circ\text{C}$	79.1 (76.8–81.2)	.756 (.732–.779)	68.5 (64.6–72.3)	.676 (.637–.713)
CA	$T_w = 0.0^\circ\text{C}$	75.5 (73.0–77.9)	.698 (.671–.725)	62.4 (58.7–66.2)	.618 (.580–.655)
CA	$T_w = 0.5^\circ\text{C}$	74.8 (72.3–77.3)	.675 (.648–.702)	60.5 (56.6–64.4)	.577 (.538–.616)
CA	Bin. logistic (refit)	73.8 (71.1–76.4)	.700 (.672–.727)	60.0 (56.1–63.9)	.598 (.558–.636)
CA	$T_w = 1.0^\circ\text{C}$	73.6 (70.9–76.2)	.645 (.617–.674)	56.8 (52.7–61.0)	.510 (.472–.550)
CA	Bin. logistic (transferred)	72.0 (69.3–74.6)	.697 (.670–.723)	59.5 (55.6–63.4)	.581 (.541–.620)
CA	$T_a = 1.5^\circ\text{C}$	64.8 (62.0–67.5)	.645 (.618–.672)	57.0 (53.0–61.1)	.514 (.475–.552)
CA	$T_a = 1.0^\circ\text{C}$	62.0 (59.2–64.8)	.620 (.592–.647)	55.1 (51.0–59.4)	.469 (.432–.508)
CO	MRoS-XGB	91.4 (88.4–94.1)	.867 (.821–.906)	82.7 (76.4–88.3)	.820 (.753–.877)
CO	$T_d = 0.5^\circ\text{C}$	87.8 (84.7–90.8)	.771 (.717–.818)	74.3 (67.2–81.3)	.692 (.614–.761)
CO	$T_w = 0.5^\circ\text{C}$	86.5 (82.7–89.9)	.763 (.706–.816)	70.4 (62.4–77.7)	.651 (.563–.732)
CO	$T_d = 0.0^\circ\text{C}$	86.3 (82.7–89.7)	.769 (.719–.816)	71.5 (64.3–78.7)	.684 (.610–.755)
CO	$T_w = 0.0^\circ\text{C}$	86.1 (82.1–89.8)	.780 (.726–.832)	69.3 (60.8–77.1)	.673 (.586–.754)
CO	$T_w = 1.0^\circ\text{C}$	85.3 (81.5–88.8)	.718 (.658–.774)	67.0 (59.1–74.9)	.559 (.473–.646)
CO	Bin. logistic (refit)	84.0 (80.1–87.7)	.677 (.618–.737)	63.7 (55.8–71.4)	.514 (.439–.593)
CO	$T_a = 1.5^\circ\text{C}$	81.3 (77.0–85.5)	.747 (.695–.796)	58.1 (50.3–66.1)	.581 (.500–.660)
CO	Bin. logistic (transferred)	80.7 (76.1–85.2)	.740 (.690–.790)	57.5 (49.5–65.8)	.575 (.492–.655)
CO	$T_a = 1.0^\circ\text{C}$	79.2 (74.6–83.6)	.739 (.687–.788)	54.2 (45.9–62.4)	.537 (.453–.618)

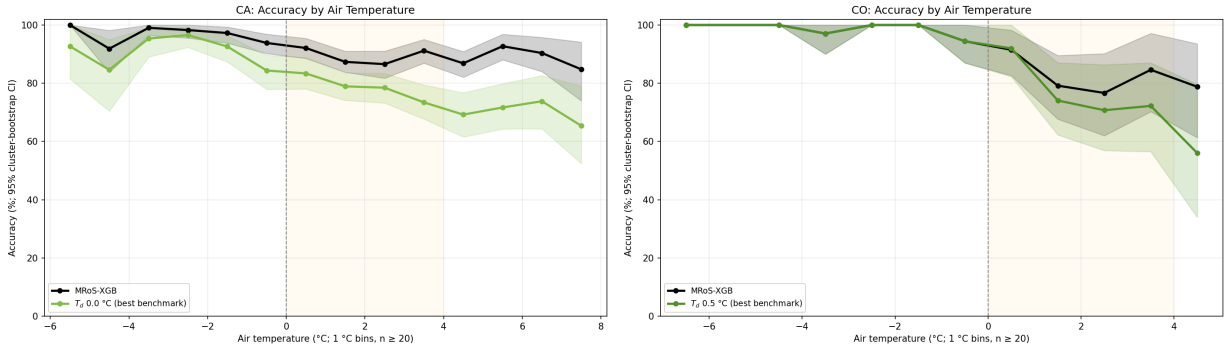


FIG. 6. Held-out accuracy by air temperature bin for MRoS-XGB (black) and the best benchmark in each domain (blue), with 95% cluster-bootstrap confidence bands, for California (left) and Colorado (right). The shaded 0°C – 4°C region marks the range of 50% rain–snow air-temperature thresholds reported across sites by Jennings et al. (2018), shown for reference; it is not the scored near-freezing subset, which is defined in wet-bulb temperature as $|T_w| \leq 2^\circ\text{C}$ (Section e). The model’s advantage grows across this range precisely as threshold methods lose skill. Accuracy is evaluated on rain and snow observations in 1°C bins plotted at midpoints, with bins holding fewer than 20 observations dropped, which is why the curves stop short of the nominal range at both tails. Ribbons are 2.5th–97.5th percentiles of 500 cluster-bootstrap resamples over 30-km $\times$ calendar-day blocks, wider than naive binomial intervals because nearby simultaneous reports are not independent. “Best benchmark” is chosen per domain and is not the same method in both panels.

melting and refreezing aloft. Satellite pLP senses the precipitating column that no surface instrument reaches and crowdsourced reports record the phase that arrived at the ground, so adding either supplies a new observation rather than a better fit to the same predictors. Interpolation makes them usable, placing every sparse source on a common grid and converting point reports into a predictor surface that

any group holding sparse phase observations could construct.

b. What crowdsourced observations contribute

The crowdsourced contribution is concentrated at the phase boundary, worth roughly 2.7 times as much near freezing as away from it (Section c), where an individual

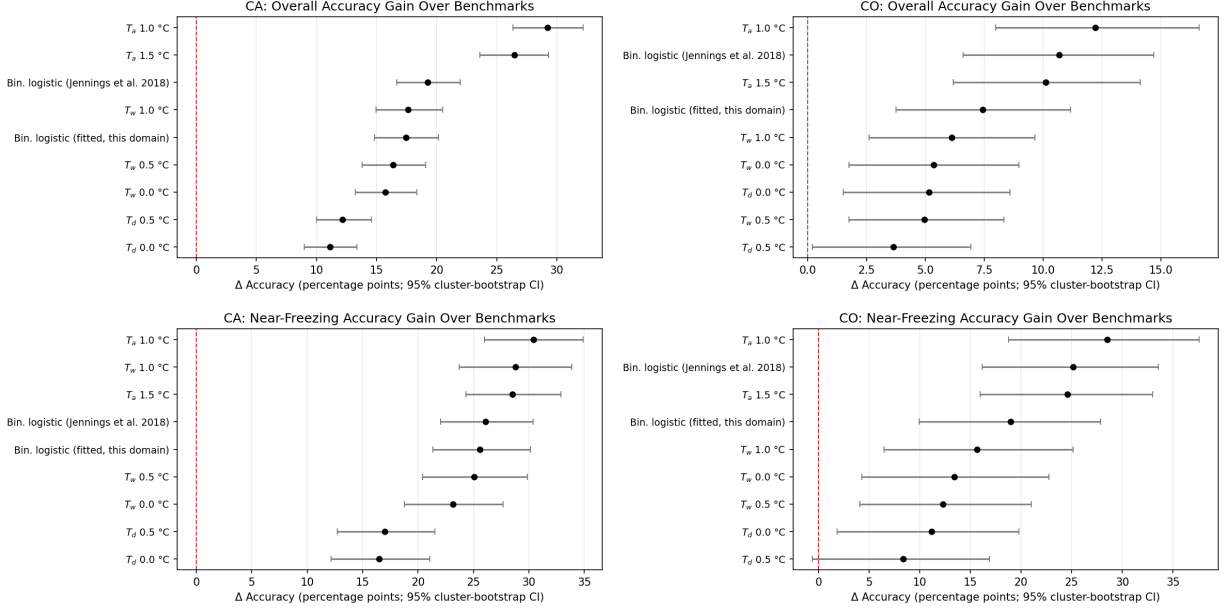


FIG. 7. Paired model-minus-benchmark accuracy differences with 95% cluster-bootstrap confidence intervals: overall (top) and near-freezing (bottom) for California (left) and Colorado (right). Every California interval lies entirely above zero. In Colorado all overall intervals do too, and only the near-freezing difference against $T_d = 0.5^\circ\text{C}$ reaches back across it. Differences are paired within each resample, so each interval reflects the variability of the *difference* on common data rather than of two independently resampled quantities. Intervals are 2.5th–97.5th percentiles of 5,000 resamples over $30\text{-km} \times \text{calendar-day}$ clusters. They quantify test-set sampling uncertainty only, propagating neither uncertainty in the kriged surfaces nor retraining variability, either of which would widen them.

observer is often the only instrument at that elevation and moment. These reports therefore record what the conventional network does not measure, rather than resampling a field the stations already describe. The relationship also runs in reverse, since a model of this type can flag reports inconsistent with their surroundings and so supply automated quality control for a data stream that is otherwise hard to validate. Which source carries the information differs by domain, with the crowdsourced surfaces dominating where the observer network is dense and satellite pLP registering instead where it is sparse (Table 4).

c. Mixed precipitation as model uncertainty

Treating mix as a region of model uncertainty rather than as a class or a discard (Section d) keeps the supervised signal binary and turns the withheld events into an out-of-sample test of model confidence. That test returns mixed evidence, since the model abstains at close to the observed mix rate in both domains without the band width having been fitted to reproduce it, yet captures only about a third of individual events. It does establish that discarded mixed reports carry usable signal, both as a predictor surface and as an evaluation set that most groups working on phase partitioning already hold.

d. Transferable methods

Two methodological results transfer beyond precipitation phase. The first is that isotonic regression, the usual recommendation for recalibrating a classifier, failed on a well-separated one in the manner described in Section (ii), while the parametric alternative could not fail in the same way. Any well-separated, class-imbalanced classifier whose probabilities will be consumed downstream should be tested for this failure rather than calibrated on the assumption that the non-parametric choice is safer.

The second concerns clustered observations, which describe most opportunistic environmental data, and where the cost of correcting for autocorrelation depends on the stage at which it is applied. Blocking the train–test split consumes training data that a dataset of this size cannot spare, whereas resampling clusters at inference only reshapes error bars on a saved prediction table (Section 3). That correction widened intervals by up to 28% without overturning a conclusion. Pairing matters equally, since comparing methods within each resample is what made the Colorado comparison reportable and is why that result can be called marginal rather than absent.

e. Relevance, limits, and outlook

The calibrated probabilities can replace static thresholds as phase forcing in hydrologic and land surface models.

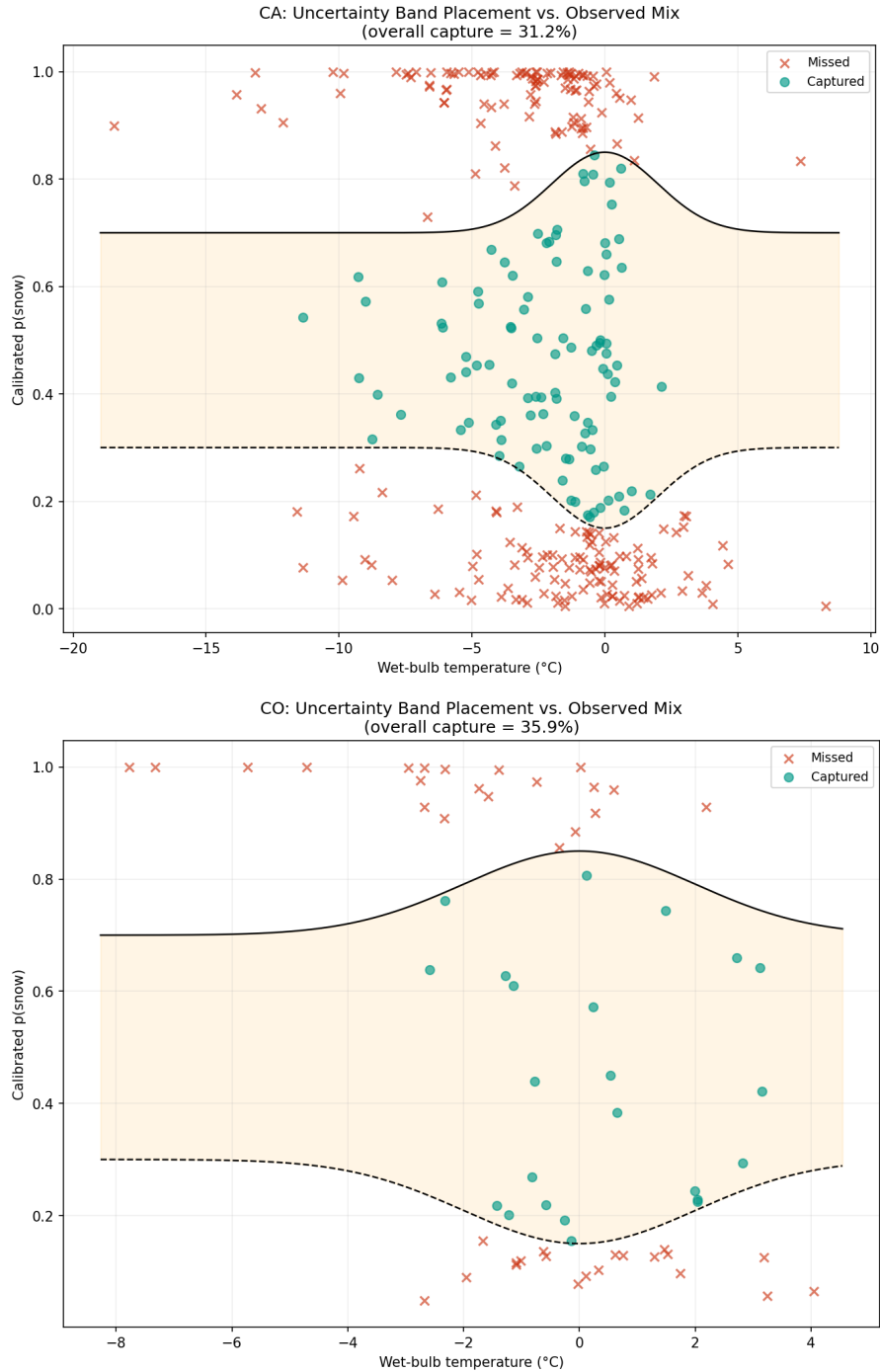


FIG. 8. The fitted uncertainty envelope and where observer-reported mixed events fall relative to it, for California (top) and Colorado (bottom), held-out test split. Solid and dashed curves are the snow and rain thresholds of Equation 1: the shaded interior is the envelope, 0.30/0.70 at both temperature extremes and opening to 0.15/0.85 at $T_w = 0^\circ\text{C}$. Each marker is one mixed event, filled where the envelope captured it and crossed where it did not. Mix was withheld from training entirely, so every marker is an out-of-sample test. The informative feature is that the misses are overwhelmingly *not* near-misses, sitting against $p(\text{snow}) \approx 0$ or ≈ 1 , so the model was confident and wrong rather than marginally short of the threshold. Widening the band would therefore recover few of them while sacrificing confident rain–snow coverage, the tradeoff quantified in Section e. Figure A10 resolves the same events by wet-bulb bin.

The uncertainty flag concentrates where phase is genuinely ambiguous, allowing an operational system to route those hours to a forecaster rather than commit to a call the model does not support. Deployment cost is bounded by the reduced-core result, since six predictors match the full eight on every overall metric and each one dropped removes an upstream dependency. That saving carries a cost near freezing, where the reduced core loses 2.4 accuracy points in California.

Whether these gains propagate to streamflow remains untested. Because partitioning choice measurably alters simulated accumulation and melt (Jennings and Molotch 2019), a gain of this size at the rain–snow boundary should register downstream, but demonstrating that it does requires forcing a hydrologic model with these fields. The contribution claimed here is therefore to phase classification rather than to streamflow prediction.

Other boundaries apply to the results themselves. The randomized split may admit mild within-storm leakage, so point estimates may be slightly optimistic for a novel season, and all intervals are conditional on the trained model and split. The MRoS network is opportunistic and skewed toward higher elevations, roads and daylight hours, observer definitions are inconsistent for mix, and a 1-km cell average cannot capture sub-grid phase transitions. With results resting on two domains and one observation network, transferability remains to be demonstrated. The clearest extension follows from the ceiling argument itself: skill near freezing stays weakest regardless of predictor set, so predictors describing the precipitating column directly, such as profile temperatures and freezing level, are the natural next addition.

7. Conclusions

We developed a framework that assimilates citizen-science phase observations, station meteorology, satellite probability of liquid precipitation, and terrain onto hourly 1-km grids, and trains a calibrated binary XGBoost classifier with an uncertainty-derived mixed-phase flag. Across two contrasting mountain domains MRoS-XGB reached a test AUROC of 0.965 in each and outperformed all nine standard partitioning benchmarks. The largest gains fall near freezing, where phase is least certain and most consequential, at 85.6% accuracy against 69.1% for the best California benchmark.

Three findings carry beyond this study. Because the ablations trace those gains to the assimilated predictors rather than to model form, the skill ceiling reported for machine learning on near-surface meteorology appears to be set by the type of observation available and is therefore relieved by new observations rather than by further work on architecture. Observer-reported mixed precipitation, which partitioning studies routinely discard, carries usable signal both as an interpolated predictor surface and

as an out-of-sample test of model confidence. And regime-stratified beta calibration and paired cluster bootstrapping both transfer to other problems of this shape.

The framework is decisive where the observer network is dense and directionally consistent where it is sparse, which leaves two questions open. A thinning experiment within one domain would establish how far observer density can fall before the crowdsourced contribution does, and forcing a hydrologic model with these fields would establish whether accuracy gained at the rain–snow boundary propagates to simulated snowpack and streamflow.

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Data availability statement. **[TODO: State where the data and code underlying the findings can be accessed, with persistent DOIs: MRoS observation archive (DOI); station data (HADS/LCD/WCC access points); GPM IMERG (GES DISC DOI); PRISM; USGS 3DEP. Code: pipeline repository (GitHub link plus archived Zenodo DOI), including the ten-stage workflow scripts and ML notebooks, with license.]**

APPENDIX

Supplementary Figures

This appendix collects supporting figures, in the order they are first cited. Each resolves a main-text result into more detail or repeats it against a second metric or binning variable as a robustness check, and none is required to follow the argument. Figure A1 gives the held-out confusion matrices and Figure A2 the reliability diagrams and calibrated score distributions. Figure A3 resolves skill against air temperature, the counterpart to Figure 4. Figure A4 gives the SHAP attribution summarized in Section c, Figure A5 presents the ablation differences with intervals, and Figure A6 repeats Figure 5 against air temperature. The remaining figures support the benchmark comparison: Figure A7 ranks the methods by accuracy, Figure A8 scores them by critical success index, and Figure A9 shows accuracy and per-phase bias against air temperature. Figure A10 closes the appendix by resolving mix capture into wet-bulb bins.

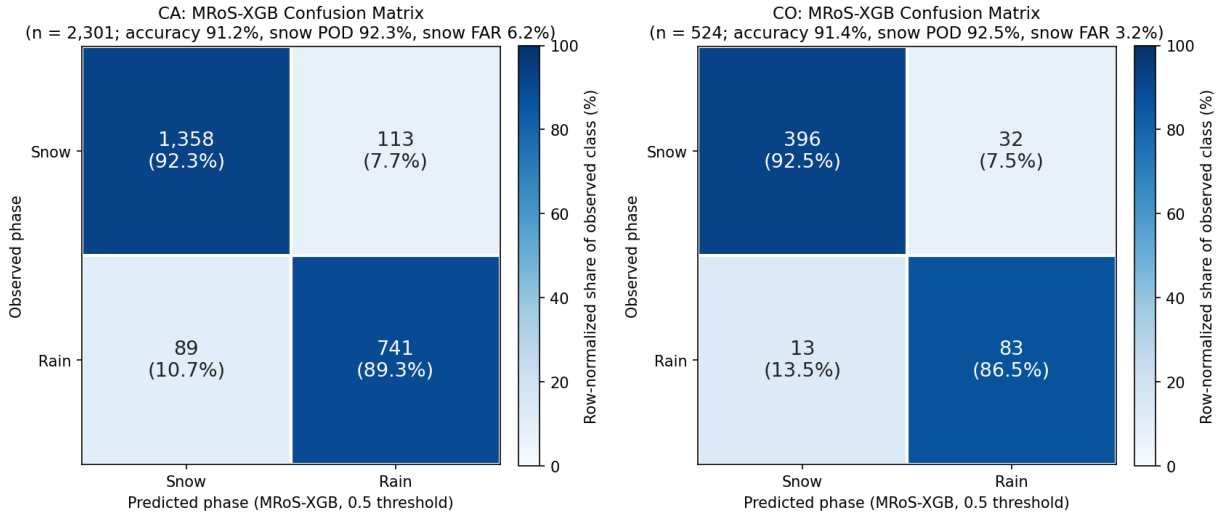


FIG. A1. Held-out test confusion matrices for California (left, $n = 2,301$) and Colorado (right, $n = 524$), scored at the raw 0.5 threshold with the uncertainty envelope not applied and observer-reported mix excluded. Cells give counts and, in parentheses, the row-normalized share of each observed class, so the diagonal entries are snow and rain recall. Panel titles report accuracy and, taking snow as the event class, the probability of detection (POD) and false alarm ratio (FAR). The two domains reach nearly the same accuracy by different routes. Colorado's snow FAR is half California's (3.2% against 6.2%) because its rain observations are few and rarely called snow, but the same scarcity leaves its rain recall lower (86.5% against 89.3%) on a base of only 96 rain observations, which is the imbalance the macro F1 and CSI comparisons of Section d are meant to expose.

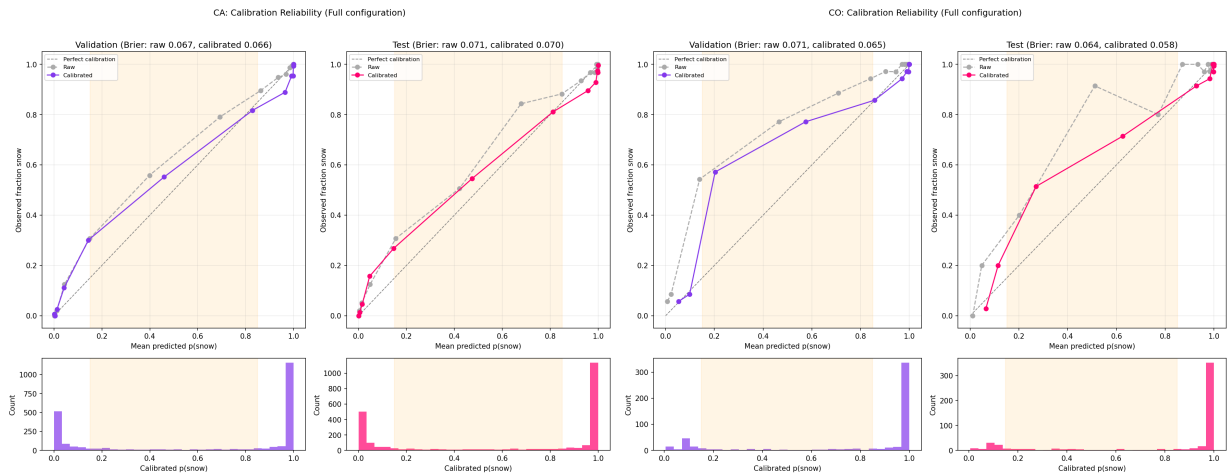


FIG. A2. Reliability diagrams and calibrated score distributions for California (left pair) and Colorado (right pair), validation and test. Beta calibration (colored) moves the raw XGBoost curve (gray) toward the diagonal; the shaded region marks the maximum extent of the uncertainty band at $T_w = 0^\circ\text{C}$ (0.15–0.85). The bimodal score histograms show that most predictions are confident, with the band occupying the sparse middle. Reliability curves use 15 quantile-spaced bins, so each point carries roughly equal sample weight, and the wide gap across the middle reflects how few predictions land there. Only pure snow and rain observations enter these diagrams, since calibration is defined against a binary outcome; mix is assessed separately in Figure 8. Brier scores in the panel titles use the same subset, raw versus calibrated. Departures from the diagonal persist in the outermost bins after calibration, for two reasons. Beta calibration is a three-parameter monotone map, so it corrects the overall shape of the reliability curve but cannot bend to fit a deviation confined to the extremes. And because the score distribution is bimodal, a quantile-spaced bin at either end spans a very narrow range of predicted probability while collecting a large share of the observations, so its mean prediction is pinned near 0 or 1 while a small number of confidently misclassified observations moves its observed frequency vertically. The residual therefore reflects cases where the model was confident and wrong, which no calibrator can remove, rather than a systematic bias in the map itself.

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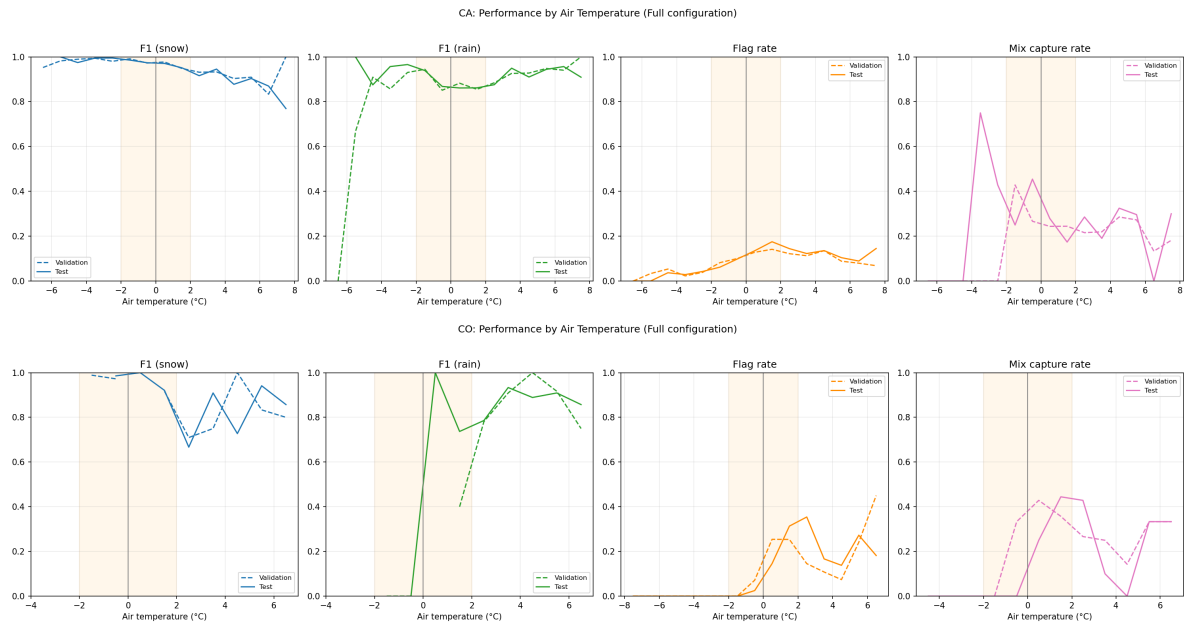


FIG. A3. Model performance in 1°C air-temperature bins for California (top) and Colorado (bottom), validation (dashed) and test (solid), with shading marking $\pm 1^{\circ}\text{C}$ about freezing. Panels show snow F1, rain F1, the flag rate, and the mix-capture rate. In this air-temperature counterpart to Figure 4, snow F1 declines as temperature rises, rain F1 improves, and the flag rate peaks just above 0°C , confirming that the envelope concentrates where phase is ambiguous under either temperature variable. Colorado's curves are noisier throughout, reflecting its smaller sample. Conventions follow Figure 4, except that bins span -8°C to 8°C because air and wet-bulb temperature occupy different ranges. A plotted zero, as in Colorado's test rain F1 near -0.5°C , means both phases were present and the model recovered no rain.

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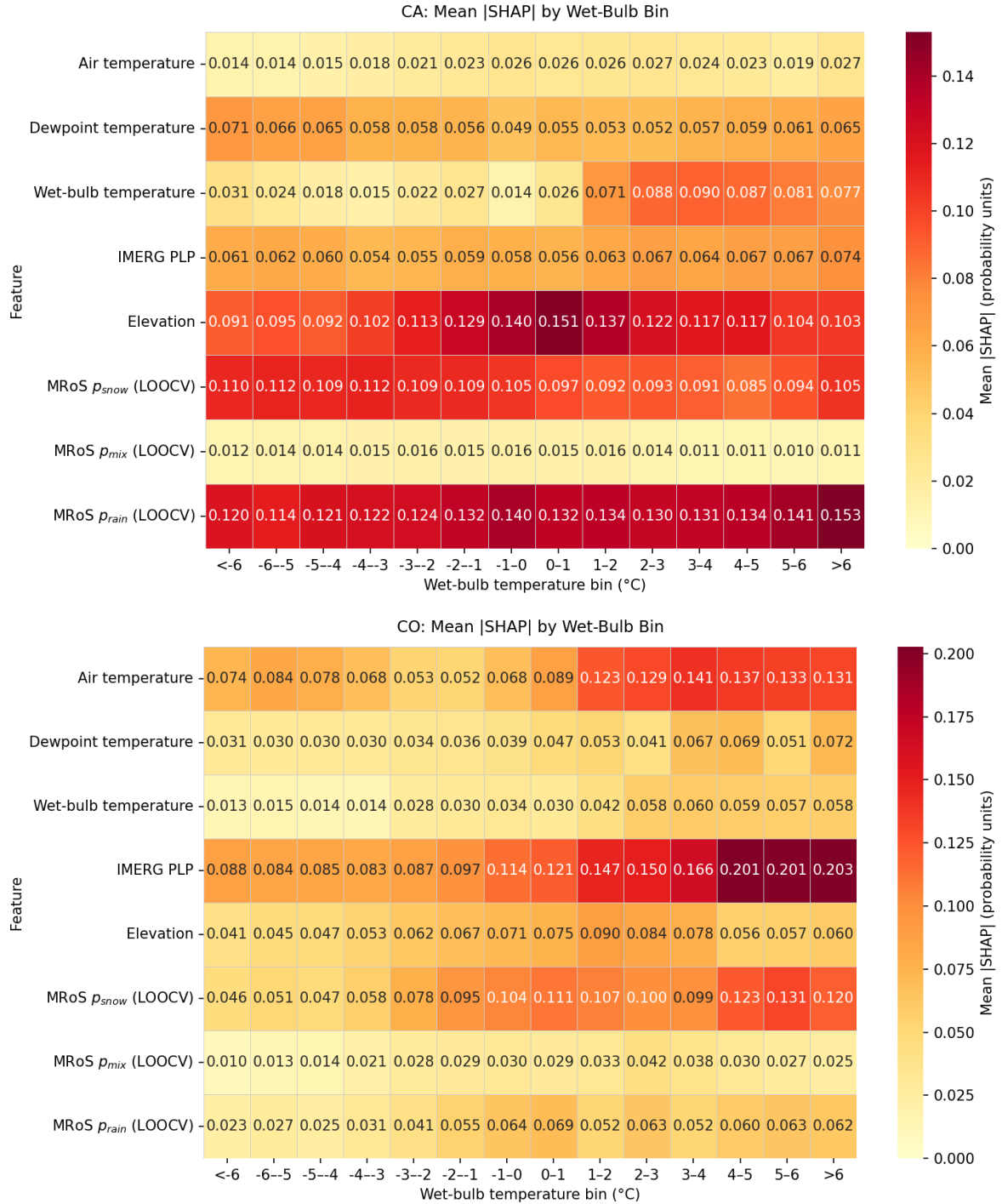


FIG. A4. Mean absolute SHAP value per predictor within 1°C wet-bulb bins, over all observations, for California (top) and Colorado (bottom). California's reliance on elevation peaks sharply in the transition zone (0.151 at $T_w = 0$ to 1°C), while Colorado's reliance on IMERG pLP climbs from 0.088 in the coldest bin to 0.203 in the warmest. As a ratio of mean |SHAP| near freezing to away from freezing, usage rises for air temperature (1.44×), elevation (1.38×) and the MRoS mix surface (1.14×), and falls for dewpoint (0.84×). Bin populations are strongly unequal, so a bright cell in an outer bin rests on far fewer observations than one near freezing, and empty bins are omitted rather than shown as zero.

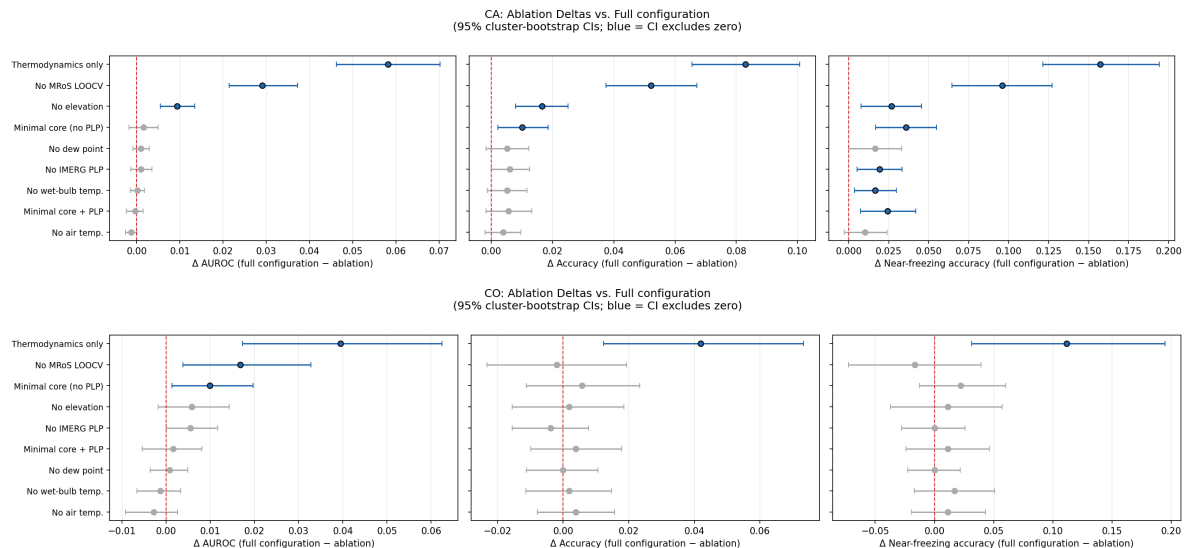


FIG. A5. Ablation differences from the full configuration with 95% cluster-bootstrap confidence intervals, for California (top) and Colorado (bottom): AUROC, overall accuracy, and near-freezing accuracy, each as baseline minus configuration, so positive values mean the removed predictors were contributing. Intervals are 2.5th–97.5th percentiles of 5,000 resamples over 30-km $\times$ calendar-day clusters, paired within resample. Markers are blue with a dark edge where the interval lies clear of zero and grey where it crosses zero, and configurations are ordered by the difference in the leftmost panel so vertical position is comparable across panels. The thermodynamics-only configuration separates cleanly from the full one on every metric in both domains, whereas most single-predictor removals stay within noise, which is the signature of predictor redundancy rather than of predictors without value. Near-freezing accuracy carries wider intervals throughout, since it is computed on the $|T_w| \leq 2^\circ\text{C}$ subset alone.

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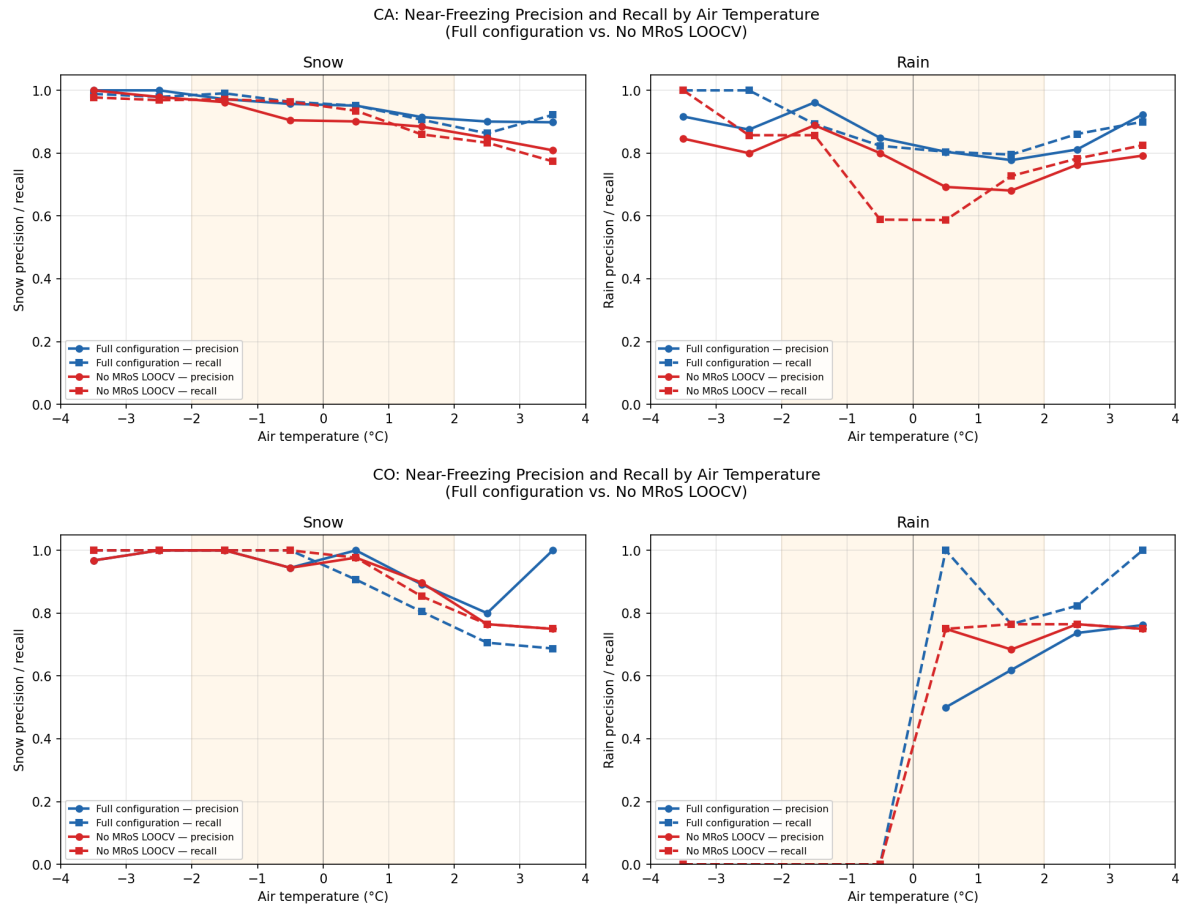


FIG. A6. Air-temperature counterpart to Figure 5: near-freezing precision and recall for the full configuration versus the configuration without the MRoS LOOCV surfaces, in 1°C air-temperature bins on the held-out test split, scored at the raw 0.5 threshold with mix excluded. In California the loss now appears in rain recall rather than snow recall, because air temperature discriminates phase less sharply than wet-bulb temperature and its bins therefore combine thermodynamic regimes that the wet-bulb view separates. That the loss survives the change of binning variable while moving between phases indicates that the crowdsourced surfaces supply information the temperature fields do not carry. Colorado's rain curves are undefined or near-degenerate in several cold bins where its rain sample is only a handful of observations.

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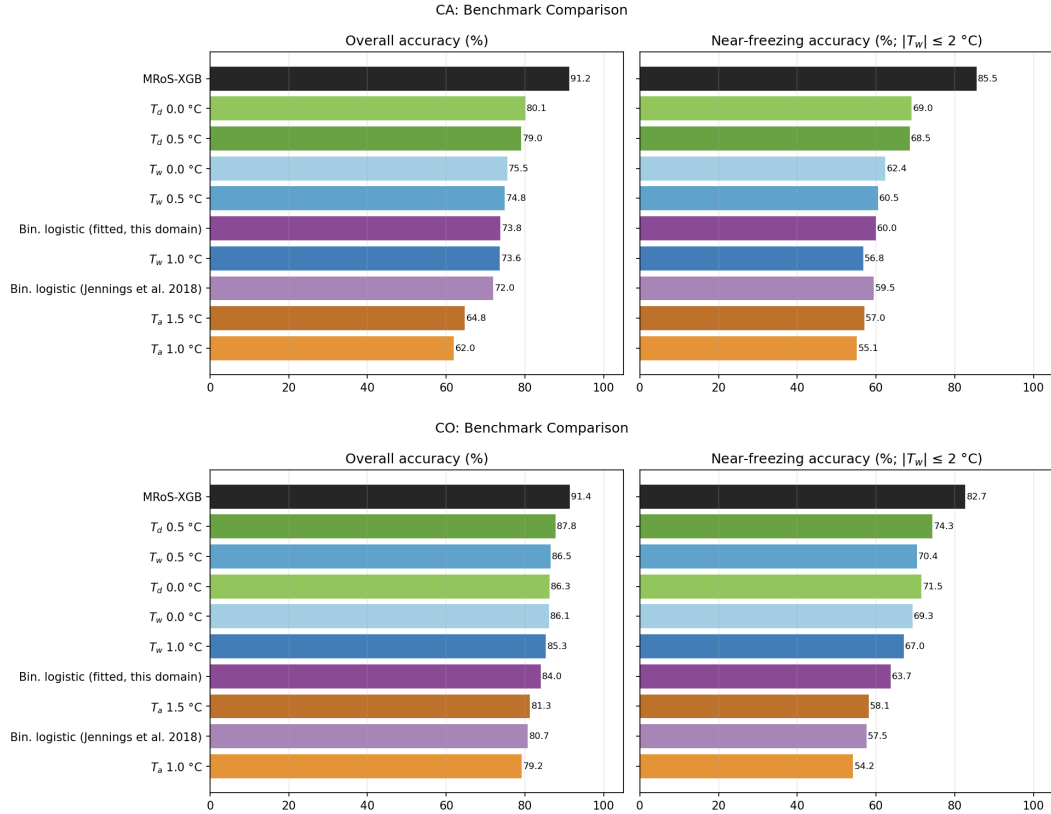


FIG. A7. Held-out accuracy of the model and the nine benchmark partitioning methods for California (top) and Colorado (bottom), overall (left) and on the near-freezing subset (right; $|T_w| \leq 2$ °C). Methods are ordered by overall accuracy within each domain and shaded by predictor family, with air-temperature thresholds in orange, dewpoint in green, wet-bulb in blue, and the two logistic methods in purple. This is the visual counterpart to Table 5, which gives the same values with confidence intervals; the paired differences that support the significance claims appear in Figure 7. Note that the near-freezing panel compresses the benchmark ordering, since every method loses skill there while the model loses less.

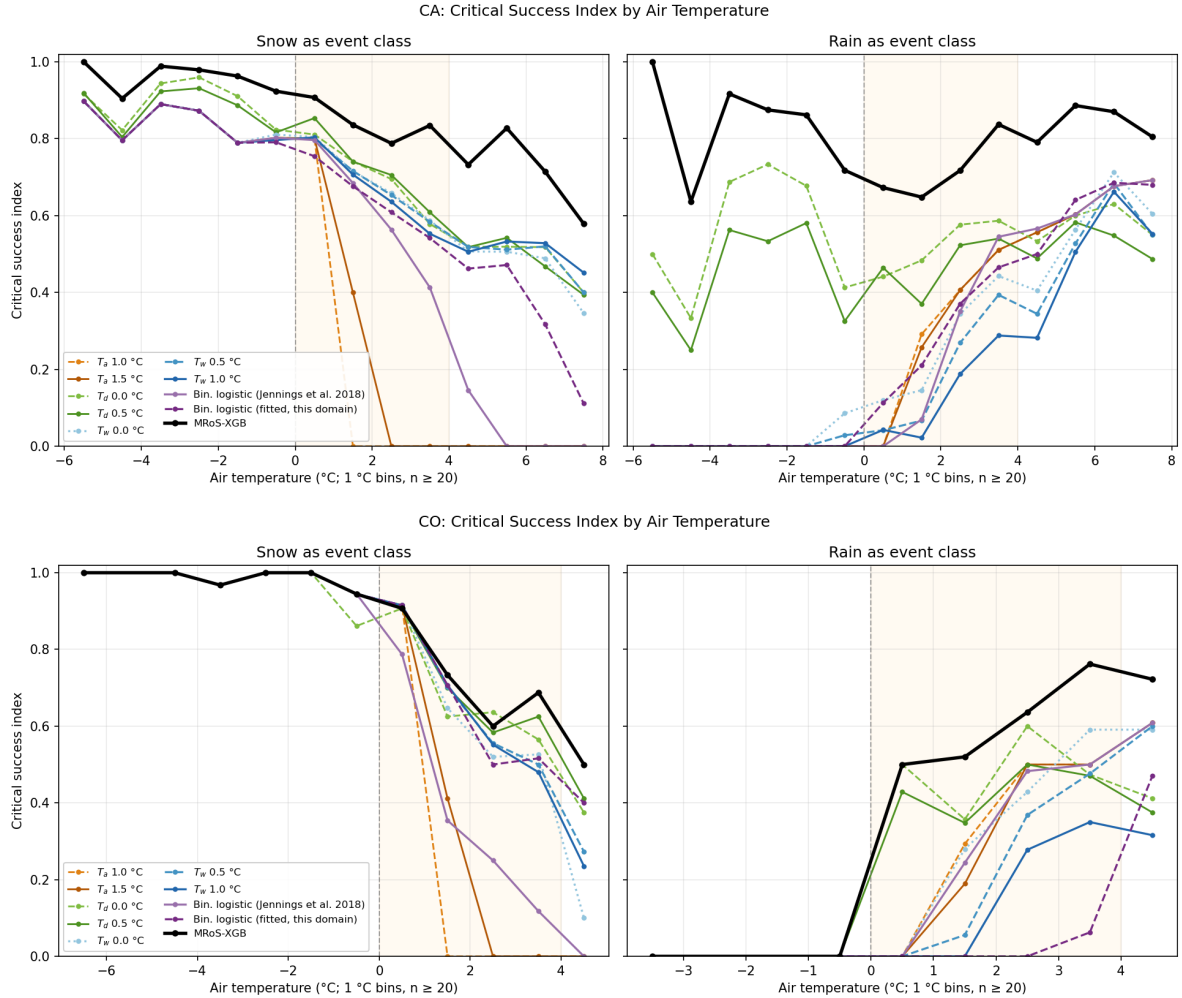


Fig. A8. Critical success index in 2°C air-temperature bins for the model and the nine benchmarks, California (top) and Colorado (bottom), scored once with snow as the event class (left) and once with rain (right). Held-out test split, with bins holding fewer than 20 pure observations dropped. As in Figure 6, the shaded 0°C–4°C region is the across-site range of published rain–snow air-temperature thresholds (Jennings et al. 2018), not the scored near-freezing subset. Because CSI excludes correct negatives, it scores each phase on its own terms and so exposes what accuracy hides in a snow-dominated sample. Every threshold method collapses toward zero CSI on the far side of its own cutoff, most visibly for the rain event class below 0°C, where a threshold that calls everything snow earns no credit at all. MRoS-XGB retains skill on both event classes across the full range. The values are not comparable between bins or between domains, since CSI depends on how often the event occurs (Table 2).

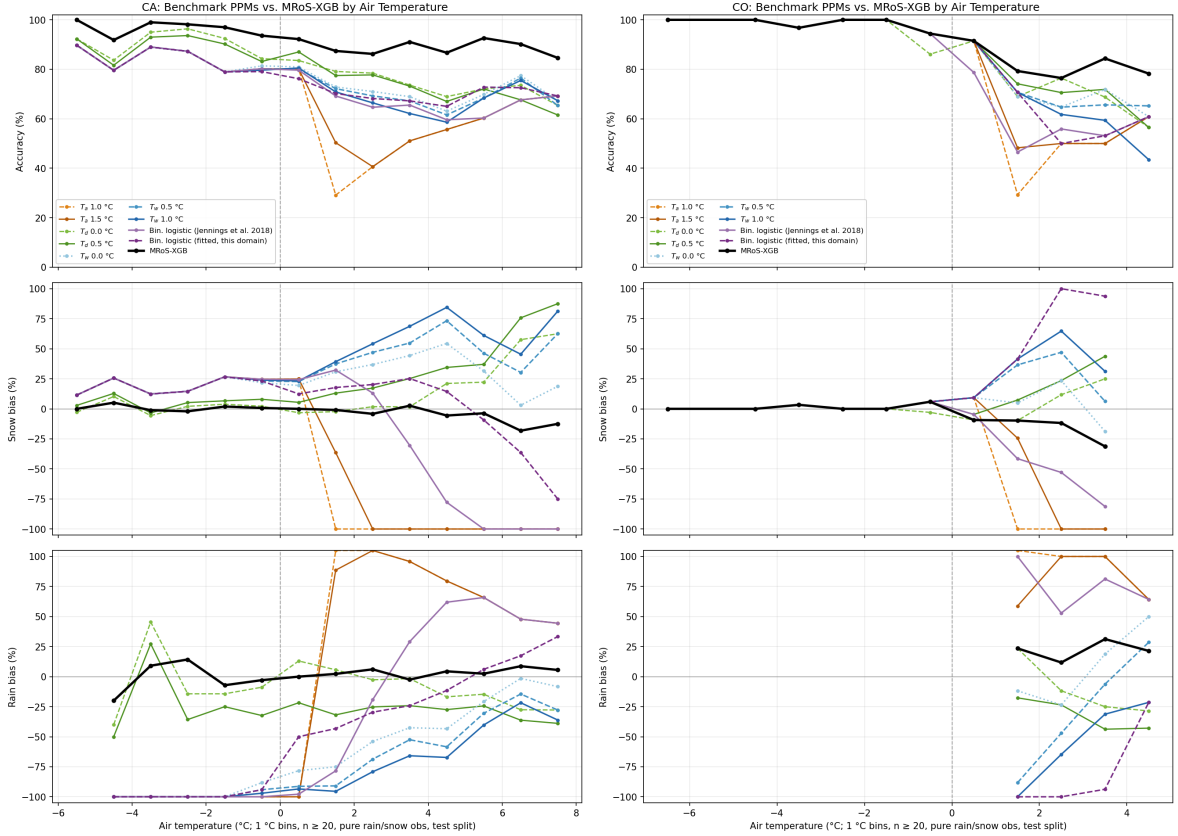


FIG. A9. Accuracy and per-phase bias against air temperature for all nine benchmark partitioning methods and the model, California (left) and Colorado (right), held-out test split, 1°C bins with $n \geq 20$ pure observations. Panels give, from top to bottom, accuracy, snow bias, and rain bias, where bias is the signed difference between predicted and observed frequency of that phase within the bin, so zero is unbiased and positive means over-prediction. These are point estimates without intervals; Figure 6 gives the bootstrap-resampled version for the model and best benchmark. Bins holding fewer than 20 pure observations are dropped, so the curves do not all span the same range.

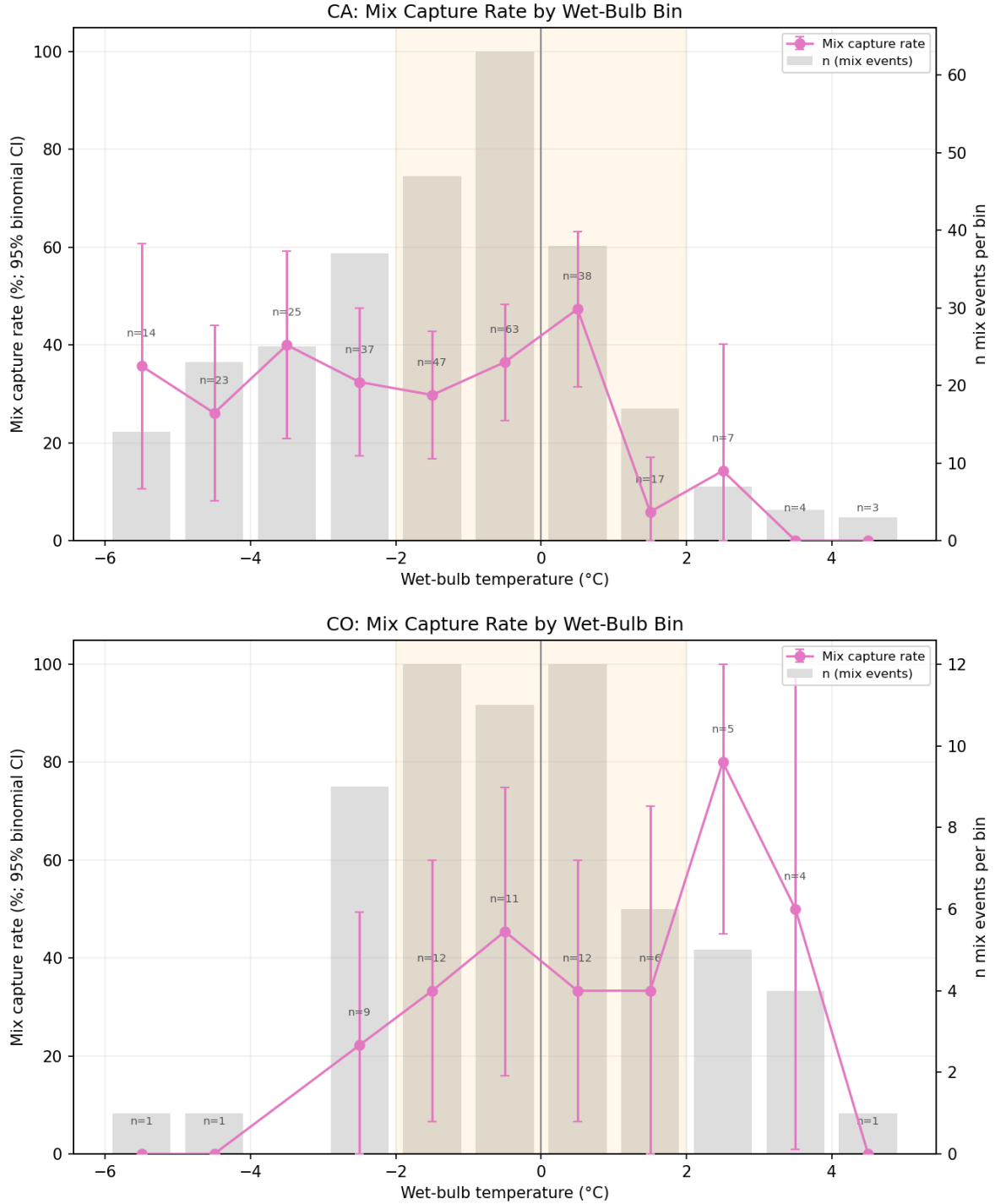


FIG. A10. Mix-capture rate by 1°C wet-bulb bin for California (top) and Colorado (bottom), held-out test split. Points give the fraction of observer-reported mix events per bin whose calibrated $p(\text{snow})$ falls inside the uncertainty band, with 95% intervals; grey bars give the mix-event count per bin on the right-hand axis, and shading marks $|T_w| \leq 1^\circ\text{C}$. Capture is highest in the near-freezing bins where mixed precipitation actually occurs (47% at $T_w = 0.5^\circ\text{C}$ in California) and falls away on either side, the intended behavior for a band placed to flag ambiguity rather than to maximize capture everywhere. Pooled capture, 31% in California and 36% in Colorado, agrees with the independent test-set bootstrap estimates of 0.30 and 0.33 in Table 3.